



DEVELOPMENT OF AN AUTOMATIC SPEECH-TO-SPEECH SYNTHESIS ARCHITECTURE FOR ENGLISH-TO-YORÙBÁ LANGUAGE USING NEURAL MACHINE TRANSLATION APPROACH

*Saturday
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Presented By

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OUTLINE

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2. **Introduction**
3. **Literature Reviewed**
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5. **Results and Discussions**
6. **Data Curation Feedback**
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1. ABSTRACT

- Different researchers have developed various tools and applications to ensure seamless communication between human beings for day-to-day transactions and interactions.
- Speech-to-Speech synthesis entails translating the speech of one language to the speech of another language.
- This proposal aims to launch the development of speech-to-speech resources using the neural machine translation approach.
- The focus of the work will be translational use case between a medical practitioner and patient as well as public health practitioner and public.
- An extension to Translatotron 2 architecture will be developed in this proposed work to translate from English to Yoruba languages.

▪ To develop the proposed model, Yorùbá-English general domain and medical domain speech corpora will be curated, and two models will be trained and developed.

▪ These are the base model and the transfer learning model using an already pre-trained English language model to another language.

▪ What follows is the training of the two models using the general domain speech corpora of Yorùbá-English curated.

▪ Second is the focus of the work, where the medical domain data will be utilised to train the two models.

▪ Thereafter comparison of the models developed under different experiments will be carried out.

▪ Finally, a prototype Softbot shall be developed and evaluated.

2. INTRODUCTION

2.1 Background

English language is the third most widely spoken native language in the world behind Mandarin Chinese and Spanish, as well as the most spoken language in the world spoken by nearly 1.5 billion speakers, which is slightly higher than 1.1 billion Mandarin Chinese speakers (Silaban et al., 2023).

Yoruba is a language that shares membership with the Niger-Congo language family and one of the three official languages in Nigeria, spoken by a population of over 50 million across South Western Nigeria, coupled with the USA, UK, Brazil, Benin and Togo (Bakenne and Salawu, 2025; Gutkin et al. 2020).

The translation of speech from one language to another is one of the research areas in speech synthesis that has gained recognition in recent years (Chang et al., 2023), and different approaches have been utilised to achieve this task.

The approaches are illustrated in Figure 1, while Figures 2 to 5 show the models available for each component of the cascaded system.

- **United Nations SDGs**
 - ✓ Goal 3 – Ensuring healthy lives and well-being
 - ✓ Goal 10 – reducing inequality within and among the community.
- **Industrial Revolution 4(IR4)**

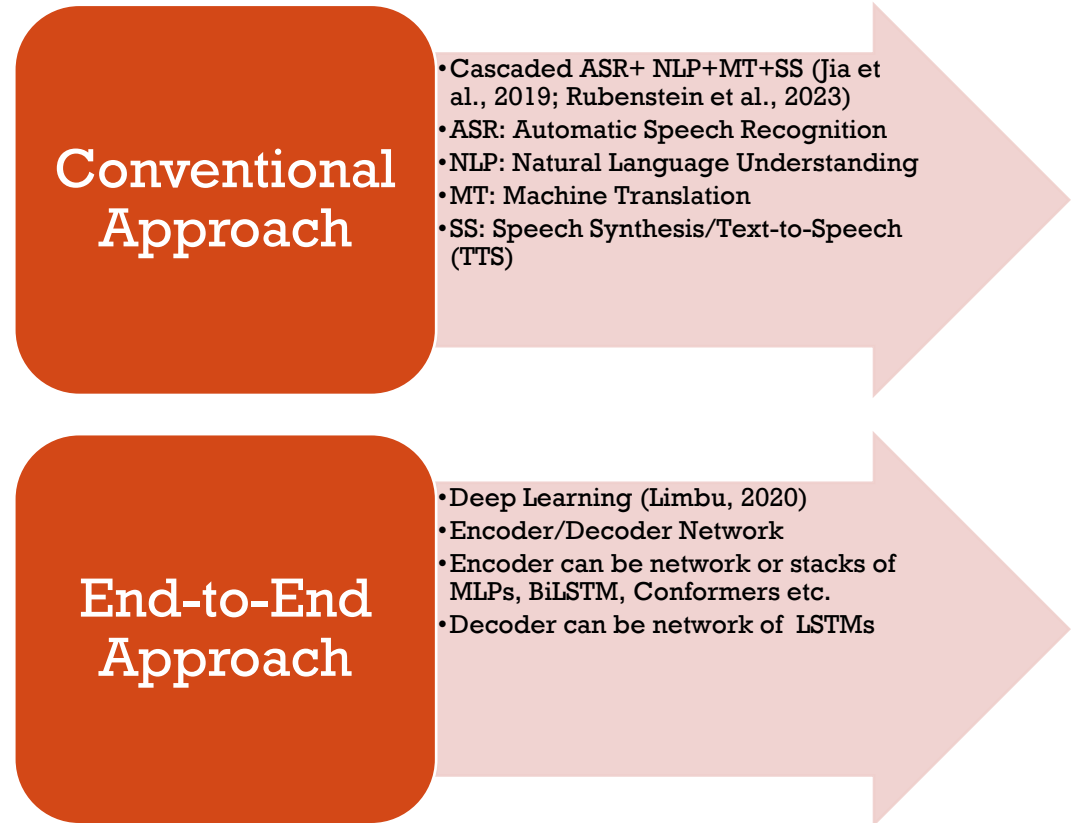


Figure 1: Approaches to Modelling Speech-to-Speech Synthesis

2. INTRODUCTION

2.1 Background (Cont.)

Well-known S2ST models developed

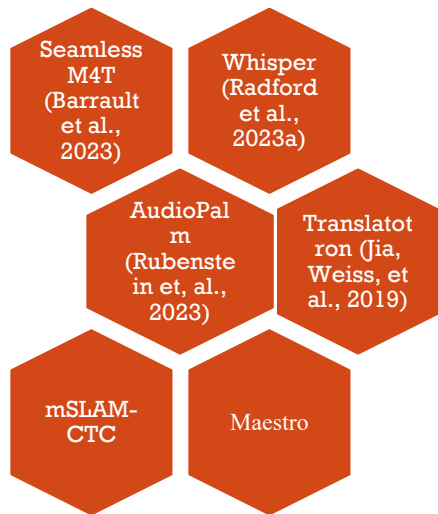


Figure 2: Well-known S2ST Models

Well-known MT models developed

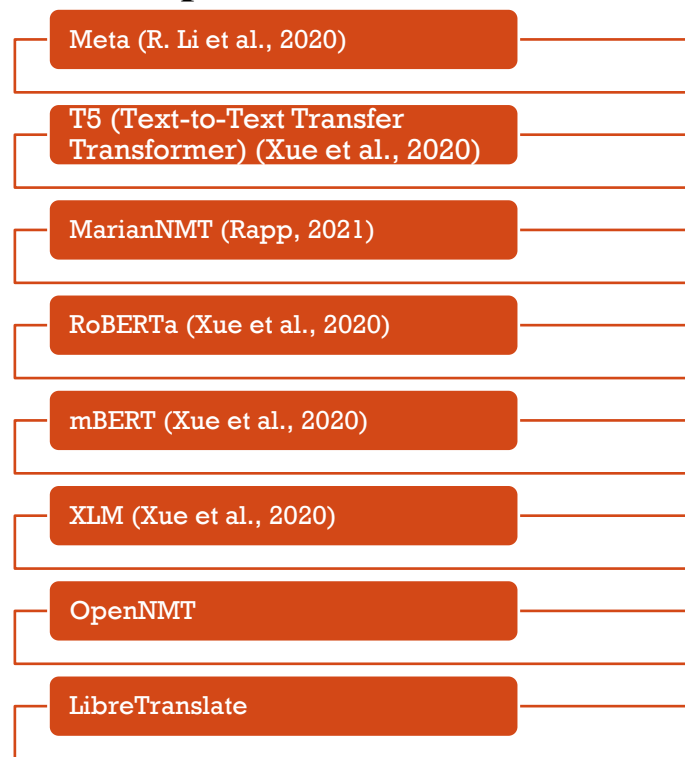


Figure 3: Well-known MT Models

Well-known TTS models developed

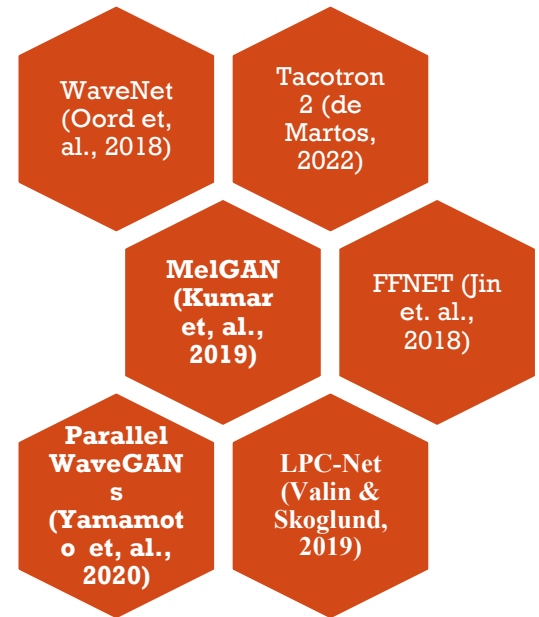


Figure 5: Well-known TTS Models

Well-known ASR models developed

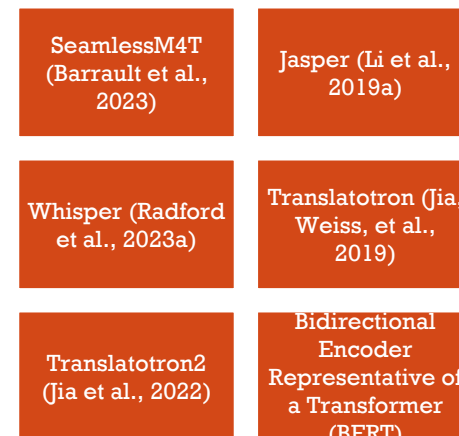


Figure 4: Well-known ASR Models



2. INTRODUCTION

2.2 Problem Statements

- The number of available speech resources or tools are not sufficient as well as Yorùbá speech corpus. (Bansal, 2019; Meyer et al., 2022) .
- The output audio samples of the target language are not real representation of the input audio samples (Limbu, 2020).
- Medical speech corpora are not available for training (Cherednichenko, 2020; Grabar & Cardon, 2019.; Qian et al., 2023).
- Language barriers (Hudelson & Chappuis, 2024; Javaid et al., 2023; Panayiotou et al., 2020; Tu et al., 2024)
- The uncertain alignment between the input and output spectrograms is challenging (Jia, Weiss, et al., 2019).
- Delays and errors are associated with the cascaded system.



2.3 Justification for the Study

- Speech-to-speech translation can minimize the language barriers experienced by individuals in their conversations with colleagues who speak other languages.
- Direct speech-to-speech translation to be considered in this work has the benefit of preserving the prosody, emotion, accent, and voice of the speaker.
- More speech data for English-Yoruba language pairs will be publicly available.

2. INTRODUCTION

2.4 Aim and Objectives

- The aim of this research is to develop an automatic Speech-to-Speech Synthesis Model for English-to-Yorùbá Language based on Neural Machine Translation Approach.
- The objectives to realise the aim are:

I. create general and public health speech corpus for English-to-Yorùba language pair with focus on the accent of speakers of the two languages.

II. preprocessing of the data collected in steps i, and ii

III. design and implement an improved architecture for speech-to-speech translation (S2ST) of English-to-Yoruba language pair.

IV. develop a prototype Softbot based on the S2ST architecture in (iii) for malaria public health education use case.

V. carry out performance evaluation of the S2ST architecture and the prototype Softbot.

2. INTRODUCTION

2.5 Scope of the Study

- Improved speech-to-speech synthesis model from English-to-Yorùbá language translation will be developed.
- The parallel speech corpora, will be curated and to be made available for research community.
- The corpora will include general and public health domains.
- A prototype Softbot based on the S2ST model will be developed

2.6 Expected Contribution to Knowledge



1. General and public health corpus for English-to-Yoruba direct S2ST corpus will be created and open-sourced for the language technology research community.
2. Improved data preprocessing algorithms for audio corpus
3. An improved direct S2ST architecture for English-to-Yoruba that can be finetuned for other low-resourced African language pairs.
4. A crowd-sourcing web application and a prototype S2ST Softbot that could be used for translation within the public health education domain and adapted for other public address translation use cases.

3. LITERATURE REVIEW

3.1 Generic Architecture of Direct S2ST

□ The architecture of a direct S2ST is illustrated in Figure 6 (Jia et al., 2019; Jia et al., 2022)

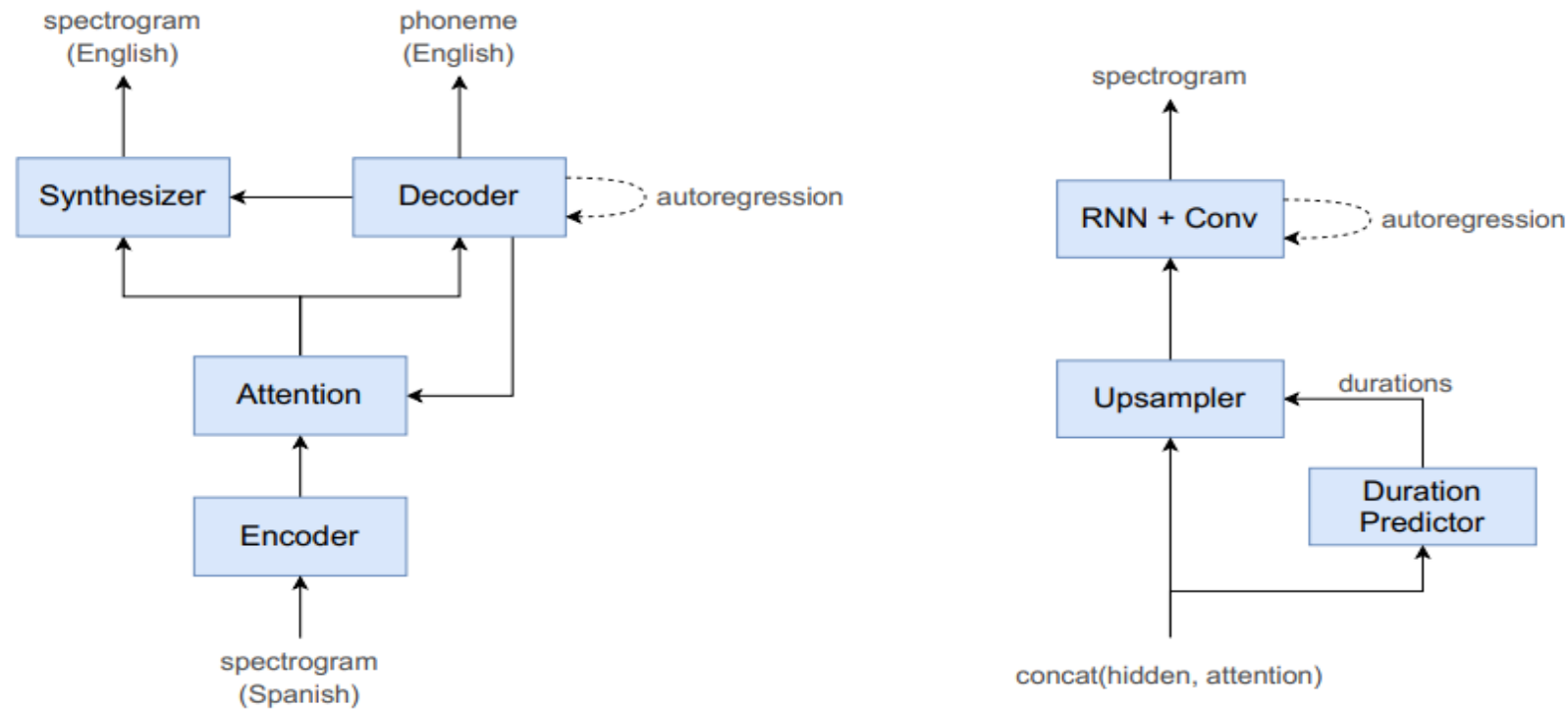


Figure 6: Generic Direct Speech-to-Speech Architecture (Translatotron 2) (Jia et al., 2022)

3. LITERATURE REVIEW

3.2 Transformer Architecture of Direct S2ST

- ❑ Model will also be developed for transformer architecture is illustrated in Figure 7 (Vaswani et al., 2017; Olaniyan et. al, 2024)

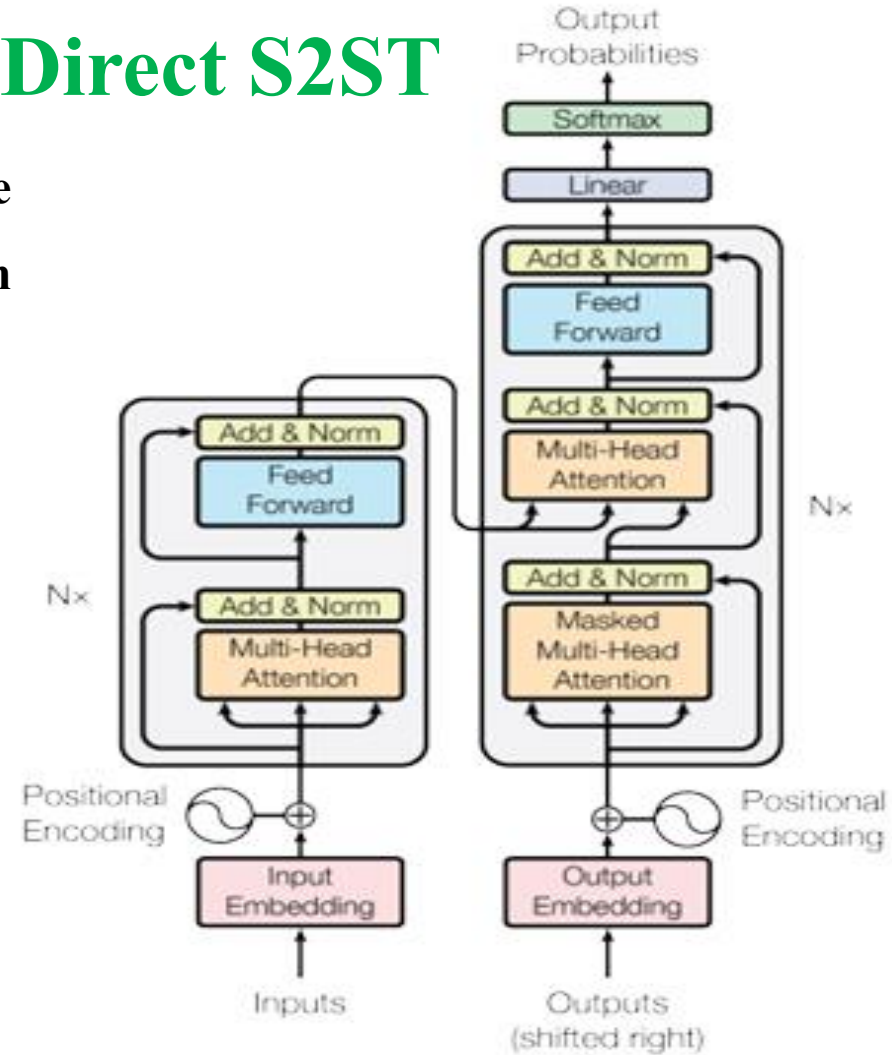


Figure 7: Transformer Architecture (Vaswani et al., 2017)

3. LITERATURE REVIEW

3.3 Summary of Literature Review



Table 1: Review of Related Works

Ref	METHODS	DATASETS/ LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Chang et al., 2023)	Sequence-to-Sequence modelling using Convolution Sub-Sampling ESPnet Toolkit for implementation	Dataset: MuST-C Average of 430 hours of audio recordings of TED talks. Languages: English-German, Spanish, French	BLEU scores using FBaank: En-De: 26.7 En-Es: 31.2 En-Fr: 37.1 BLEU scores using Discrete tokens: En-De: 28.6 En-Es: 33.0 En-Fr: 38.7	(1) Use of discretisation for speech extracted features. (2) Use of De-duplication and sub-word modelling to further reduce the size of the speech features.	Advantages 1. Performance enhancement 2. Reduction in training time due to discrete token. Limitations 1. The work did not consider the comparison of other discretisation approach

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Jia et al., 2019)	Attention Sequence-to-Sequence approach with: Speaker Encoder, and Neural Vocoder Translatotron	Datasets: Conversational Spanish-to-English dataset: 979k parallel utterance pairs Fisher Spanish-to-English dataset: Spanish-English	Conversational Spanish-to-English dataset: BLEU scores: Source + Target model: 42.7, Cascaded model (STT & TTS): 48.7, Ground truth: 74.7 Fisher Spanish-to-English	Mel-spectrogram (MFCCs)	Advantages 1. It explore different model trainings. 2. It utilises multi-head attention for speech-to-speech translation. Limitation 1. The computational time of the algorithm was not considered.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/LANGUAGES	RESULTS	FEATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Jia et al., 2022)	Direct speech-to-speech Translatotron 2	Datasets: Conversational es→en, Fisher es→ en, Multilingual CoVoST2 (es, fr, de, ca→en) Languages: Spanish-English, French-English, German-English, Catalan-English	Translation Quality Fisher dataset: outperformed Translatotron by 15.5 BLEU score. Subjective MOS Naturalness Fisher dataset: Translatotron: 3.70 ± 0.08 Translatotron2: 3.98 ± 0.08	Mel-spectrogram features	Problems it addresses from the original Translatotron: 1. Non-contribution of the attention alignment learned. Advantages 1. Performance was closer to the cascaded system, Limitations 1. it does not consider the discrete rep.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTI ON METHODS	ADVANTAGES / LIMITATIONS
(Olaniyan et. al., 2024)	Direct speech-to-speech (Transformer architecture) Augmented Attention Mechanism (73 hours / 71 hours)	Datasets: 200,023 audio recordings Source: Primary Public domain Language: English-Yoruba	BLEU SCORE: 0.7, MOS: 2.8, Translation Accuracy: 98 Original Transformer translation Accuracy: 88.16%	Spectrogram based model	Advantages: 1. Augmented attention mechanism to capture speech representation such as the diacritics, prosody, intonation etc. 2. More attention head (42) utilized for training. Limitations: 1. The dataset could be scaled up for enhanced performance. 2. training input data is spectrogram based. 3. A trade-off between choice of attention-head and complexity.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Prabhjot et. Al., 2024)	Sequence-to-sequence model: Direct Unit-to-Unit Translation (U2UT) model	Datasets: 1. FLEURS with 1625 parallel Punjabi-English speech. 2. Synthetically generated English speech Language: Punjabi-English	U2UT Model: BLEU score: 3.9829 S2UT Model: BLEU score: 0.2954 Cascaded Model (Whisper-large-v3): BLEU score: 18.1136	U2UT Model: No feature extraction was utilised. HuBERT Base model trained on 960 hours of Librispeech	Advantages: 1. The use of discrete speech units that reduces the dimension of the speech signal while also preserving the original Limitations: 1. The size of the dataset was not sufficient enough for training; hence performance of the proposed approach was lower than the cascaded system implemented with Whisper-large-v3.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/LANGUAGE	RESULTS	FEATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Radford et al., 2023)	Transformer encoder & decoder using MLP AdamW optimiser	LibriSpeech, TED-LIUM 3, Common Voice 5.1, Artie bias Corpus, Call Home & Switch board, WSJ, CORAAL, CHiME-6, AMI-IHM AMI-SDM1, Meanwhile, Rev16, Kincaid 46 Earnings-21, etc. LANGUAGE: Non specific, multilingual ASR	Transcription Multilingual LibriSpeech: Whisper large-v2 (Spanish): WER = 4.2 %, Common Voice 9: Whisper large-v2 Dutch: WER = 5.8 % Spanish: WER = 5.6 % FLEURS:	GELU activation function, MFCC	ADVANTAGES 1. Robust to noise (additive noise & pub noise) LIMITATIONS 1. Ideal robustness could not be achieved. 2. Performance varied across different dataset.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Rubenstein et al., 2023)	Discretisation + Decoder-only transformer model Models: AudioPaLM 8B AST, AudioPaLM 8B S2ST, AudioPaLM-2 8B AST,	Datasets: CoVoST2, CVSS, VoxPopuli, Common Voice, etc. Language: Nonspecific	CoVost2 Dataset (AST Task): BLEU scores: , AudioPaLM-2 8B AST: 37.8, AudioPaLM-2 8B cascaded ASR + translation: 39.0 Whisper Large-v2: 29.1	No feature extraction Discrete tokenisation	Advantages: 1. It performed better than the SOTA Whisper-Large-v2 Model. Limitations: 1. It is not a sequence-to-sequence modelling, 2. There might be possible delays introduced during training.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Olatunji et al., 2023)	Collection of texts / transcripts from sites such as PubMed, NCBI, wiki-texts, developme nt of audio recording web page.	Datasets: AfriSpeech Language: English language of African accent from 13 countries.	AfriSpeech-clinical data Whisper medium: WER of 0.568 Wave2Vec large: WER of 0.415 LibriSpeech(test-clean) Whisper large pre-trained model: WER of 0.167	Nil	Advantages: 1. Accented corpus. 2. Curated for general and medical domain. Limitations: 1. Not curated for African indigenous languages.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Anmol Gulati et al., 2020)	Conformer model for ASR.	Datasets: LibriSpeech Language: English language.	Big Conformer model with 118 million model parameters without a LM: WER = 0.021 With LM: WER = 0.019	Spectrogram-based	Advantages: 1. Ability to capture long term dependencies (global) by transformer and local relationship by th help of CNN Limitations 1. Just LSTM decoder was utilized for the end-to-end ASR model developed.

3.4 Gaps in the Literature

Table 2: Summary of Gaps in the Literature

Consideration	Authors	Gaps
Dataset	(Bansal, 2019); (Kaur et al., 2024); (Cherednichenko, 2020); (Grabar & Cardon, 2019); (Qian et al., 2023); (Kaur et al., 2024)	Insufficient amount of medical and nonmedical parallel speech corpus for the modelling of speech-to-speech translation Difficulty in acquiring parallel speech corpus
Complexity of the architectures	(Jia et al., 2022)	The system is complex due to the architecture implemented, and the computational complexity of the models is not mostly stated

3.4 Gaps in Literature

Table 2: Summary of Gaps in Literature (Cont.)

Consideration	Authors	Gaps
Speech discretisation	(Chang et al., 2023) (Rubenstein, Asawaroengchai, Nguyen, Bapna, Borsos, Quitry, Chen, Badawy, Han, Kharitonov, & others, 2023).	Limitation on work done Possible generation of errors due to lack of discretisation
Cross-lingual voice transfer	(Jia, Weiss, et al., 2019); (Kaur et al., 2024)	Challenges are encountered with cross-language voice transfer, and performance was lesser than the cascaded system

3. LITERATURE REVIEW (CONT.)

3.4 Gaps in Literature

Table 2: Summary of Gaps in Literature (Cont.)

Consideration	Authors	Gaps
Complexity of the architectures	(Jia et al., 2022)	The system is complex due to the architecture implemented, and the computational complexity of the models is not mostly stated

4. METHODOLOGY

Table 3: Table of Methodology to Achieve Specific Objectives

Specific Objectives	Methodology	Data	Expected Results
1	State-of-the-art text-to-text translation tools, Recording of the parallel corporal of both languages using a user friendly web site, Use of SPSS/MATLAB for analysis.	English text and corresponding Yorùbá text, Crowdsourced Human beings.	Quality accented English-Yorùbá parallel corpus
2	Sampling, Feature extraction: MFCCs, Discretisation: Hubert Modelling, K-means clustering,	Outputs from Objective 1	Discretised features with reduced dimension ready to be fed as input to the model.

4. METHODOLOGY

Table 3: Table of Methodology to Achieve Specific Objectives

Specific Objectives	Methodology	Data	Expected Results
3	Sequence-to-modelling using transformer architecture, and training	Outputs from Objective 2	State-of-the-art English-Yorùbá language translation model with
4	Utilisation of various software packages like the Python, HTML,CSS, APIs etc.	Output from Objective 3	User friendly Softbots language translator.
5	Computation of BLEU scores, and MOS Naturalness and MOS Similarity, as well as testing of Softbot by selected individual	Outputs from Objective 3 and 4	SOTA BLEU scores and MOS that is typical of a S2ST models.

4. METHODOLOGY

4.1 Proposed Architecture

- Figure 8 is the architecture of this proposed study to provide a broader understanding of the work.
- An improvement on Translatotron 2 (Figure 6)

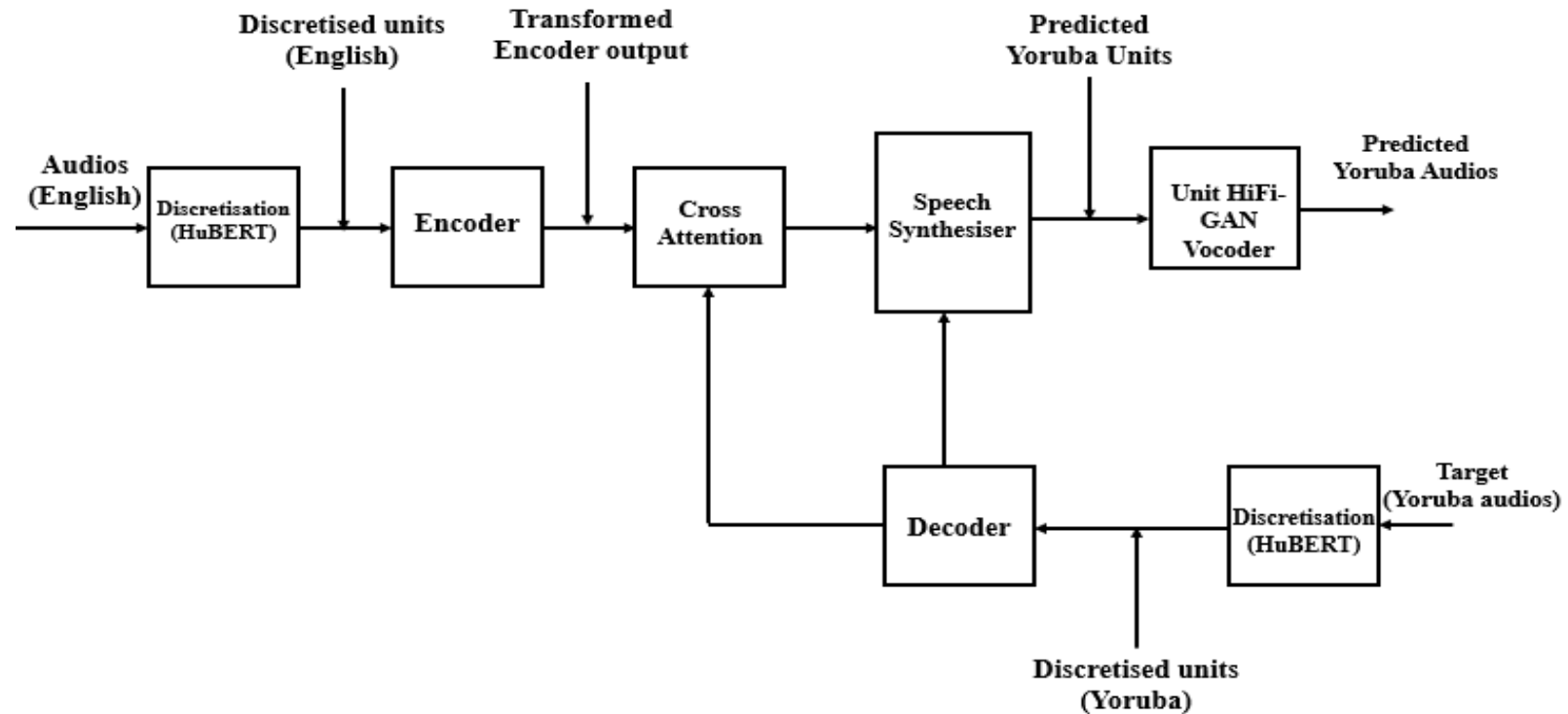


Figure 8:The Proposed U2UT (S2ST) Neural Machine Translation Architecture

4.1 Proposed Architecture

4.1.1 Comparison of Models

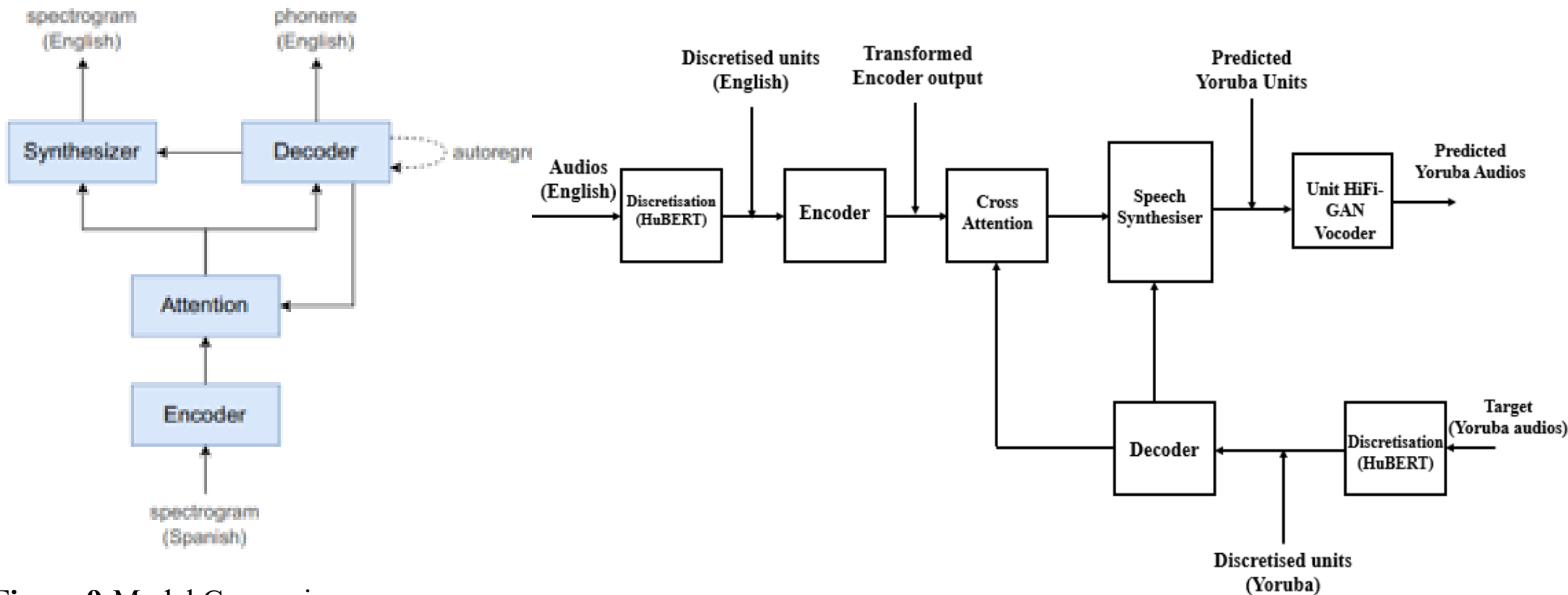


Figure 9:Model Comparison

4.1 Proposed Architecture (Cont.)

- The speech synthesizer of U2UT Model is shown in Figure 10.
- It is an extension to Translatotron 2, a SOTA speech-to-speech model.

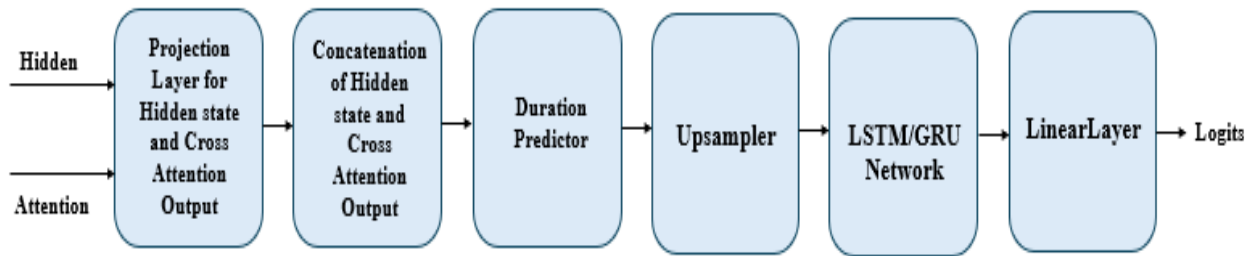


Figure 10:The Speech Synthesizer of U2UT model

3.1.1 Major Extension to be carried out

- English -Yoruba language pair corporal to be utilised for training.
- Speech discretization to be utilised unlike MFCC features, for both source and target languages.
- Stacks of LSTM, BiLSTMs shall be consider for the encoder and decoder.
- LSTM, BiLSTM shall be considered for the decoder of the speech synthesizer along with convolutional network.



4. METHODOLOGY

4.1 Proposed Architecture (Cont.)



Table 4: Translatotron2 versus Proposed Architecture

S/N	Feature	Translatotron2	Remodified Translatotron2 U2UT
1	Input Representation	Source spectrogram	Source discrete units
2	Target Representation	Target phonemes (for alignment) + target text + target audio	Target discrete units only (captures phonetics, prosody, intonation)
3	Alignment (Cross Attention)	Between source spectrogram and target phonemes (to avoid instability of spectrogram-to-spectrogram)	Between source discrete units and target discrete units (simpler, more stable, retains speech features)
4	Output	Target audio + target text	Target audio only (phoneme output not required)
5	Vocoder	Spectrogram-based vocoder (e.g., Griffin-Lim, WavRNN)	Unit-based vocoder (Unit HiFi-GAN) finetuned on Yoruba discrete units–audio pairs
6	Languages Used	Spanish→English, German→English, Catalan→English (all high-resource)	English→Yoruba (low-resource language)
7	Encoder	Conformer encoder	Conformer encoder (same as original)
8	Synthesizer	LSTM-based	Both LSTM and GRU tested
9	Training Complexity	Higher (spectrograms are continuous, high-dimensional)	Lower (discrete units reduce complexity, errors, and training instability)
10	Speech Feature Retention	Phonemes capture linguistic features but lose prosody/intonation	Discrete units retain prosody, intonation, pitch, and voice transfer
11	Training Stability	Risk of instability (spectrogram-to-spectrogram alignment unstable)	More stable, faster convergence, fewer errors (confirmed in ablation studies)

4.2 Architectural Design

4.2.1 HuBERT Versus Other Discrete Unit Generation Models Experiments

Units to audio generation experiments were carried out involving the comparison of speech generation using discrete units of HuBERT, Wav2vec, and WavLM.

4.2.2 Evaluation of the Discrete Units plus Audios Experiments

The evaluations of the audios were carried out using STOI, PESQ, and DNSMOS.

4.2.3 Choice of Unit based Vocoder for S2ST Model

Finetuned HuBERT based unit HiFi-GAN vocoder (pretrained on the medical domain containing 1000 Yoruba audios of the GC-ENYOR) was integrated into the U2UT model.

4.2.4 Design of the U2UT model



4. METHODOLOGY



4.2 Architectural Design

4.2.4 Design of the U2UT model

A. Python Framework

For the experiments carried out in this project, the Pytorch framework was utilised. It provides the necessary modules with any class name that can be utilised for seamless for model development and training.

B. Creation of Virtual Environment and Importing of Python Module

The virtual environment, which is a dedicated and isolated environment from other model developments was created for the Python IDLE on Ubuntu.

Python 3.12 was utilised for U2UT model development and unit-audio HiFi-GAN vocoder experiment, while Python 3.13 was utilised for discrete unit extraction from HuBERT, Wav2vec, and WavLM pretrained models.

C. Audio Preprocessing / Augmentation Techniques

1. Resampling
2. Silence Trimming
3. Speed Perturbation
4. Pitch Shift
5. Noise Addition
6. Data Augmentation

4.2 Architectural Design

4.2.4 Design of the U2UT model

D. HuBERT Discretisation

- The HuBERT features were first extracted from the audios using the `Wav2Vec2FeatureExtractor()` method of the transformers module, and the output of this was fed to the `HubertModel()` method (also of the transformers module), and continuous HuBERT values were obtained (last hidden state).
- Discrete values were obtained using the `joblib.load()` module, where the argument was set to `hubert_base_ls960_L9_km500.bin`, which was downloaded from the repository and saved into the project directory.
- The output of this were discrete values with shape `[number_of_audios, sequence_length]`.

E. Mapping of English Discretised Values to Embedding Space

The new shape becomes `[number_of_audios, sequence_length, d_model]`

4. METHODOLOGY

4.2 Architectural Design

4.2.4 Design of the U2UT model

F. Feedforward Module

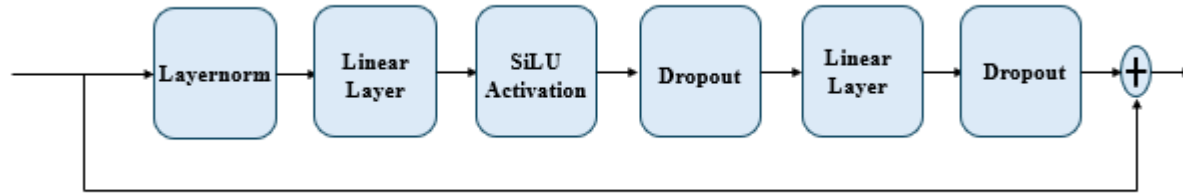


Figure 11: Feed Forward Module

G. Multi-Head Self Attention (MHSA)



Figure 12: Multi-Head Self Attention Module

H. Conformer Convolutional Module

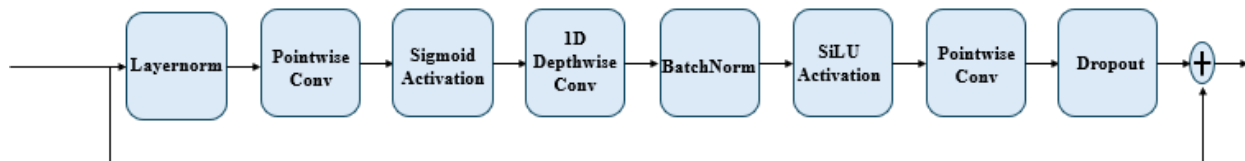


Figure 13: Convolutional Module

I. Encoder and Block of Encoders Module

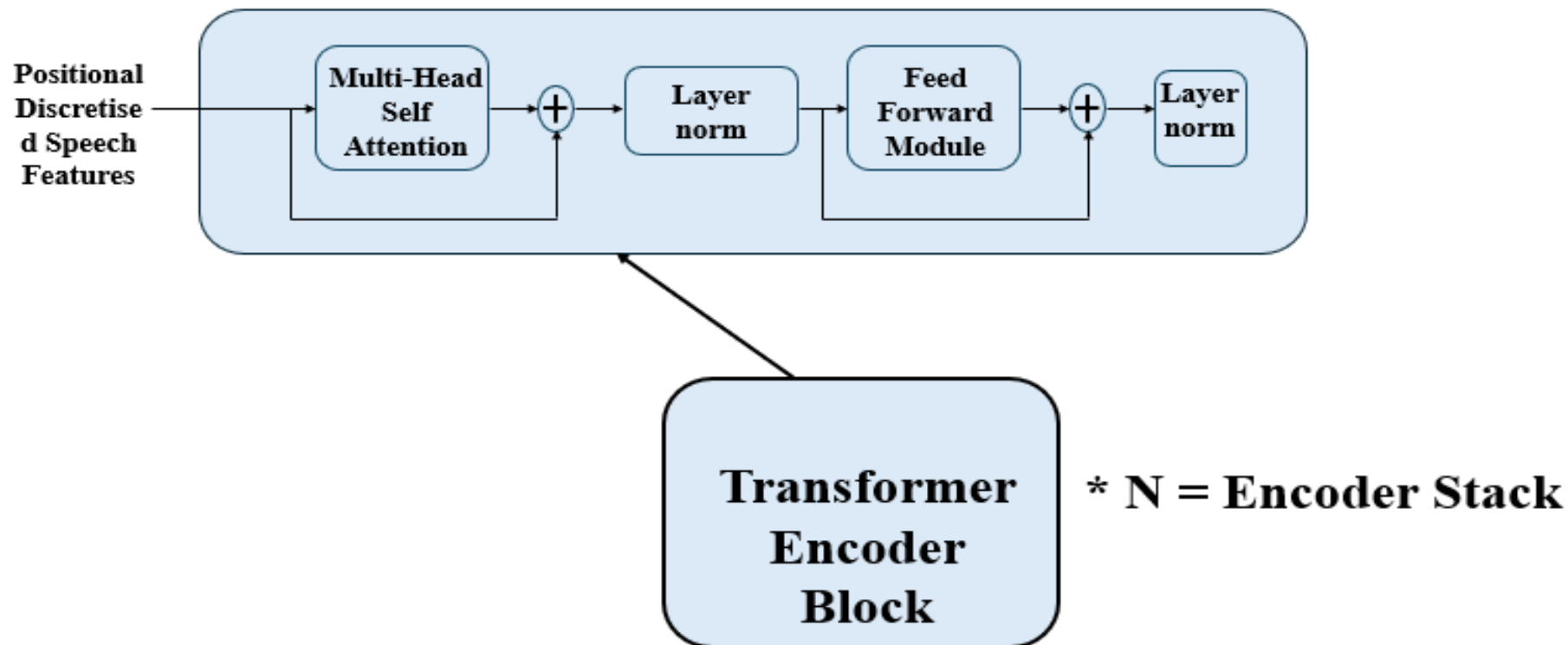


Figure 14: Transformer Encoder

4.2.4 Design of the U2UT model

I. Encoder and Block of Encoders

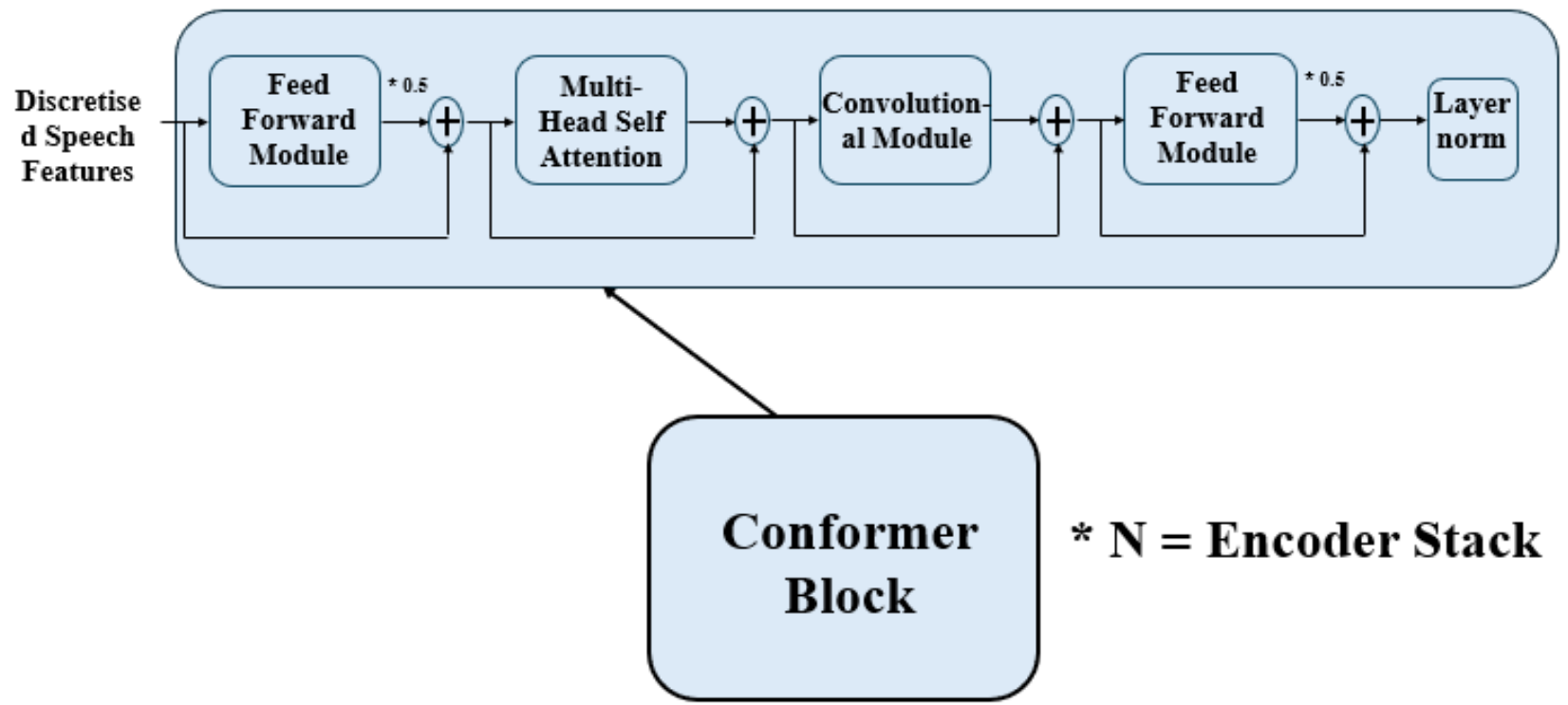


Figure 15: Conformer Block

4. METHODOLOGY

4.2 Architectural Design

4.2.4 Design of the U2UT model

I. Encoder and Block of Encoders

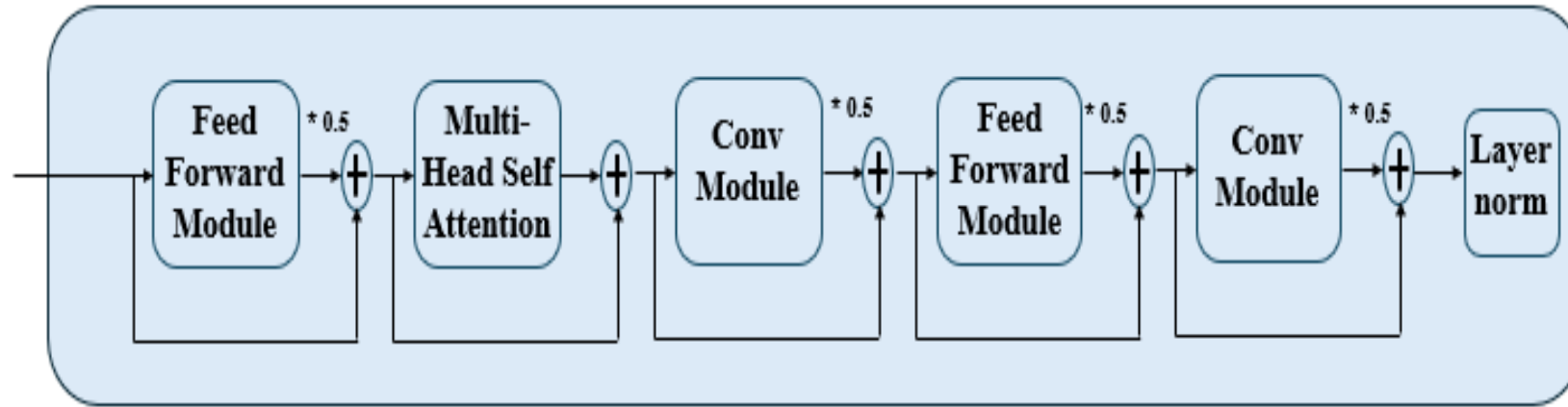


Figure 16: Conformer Block with Altered Convolutional Module

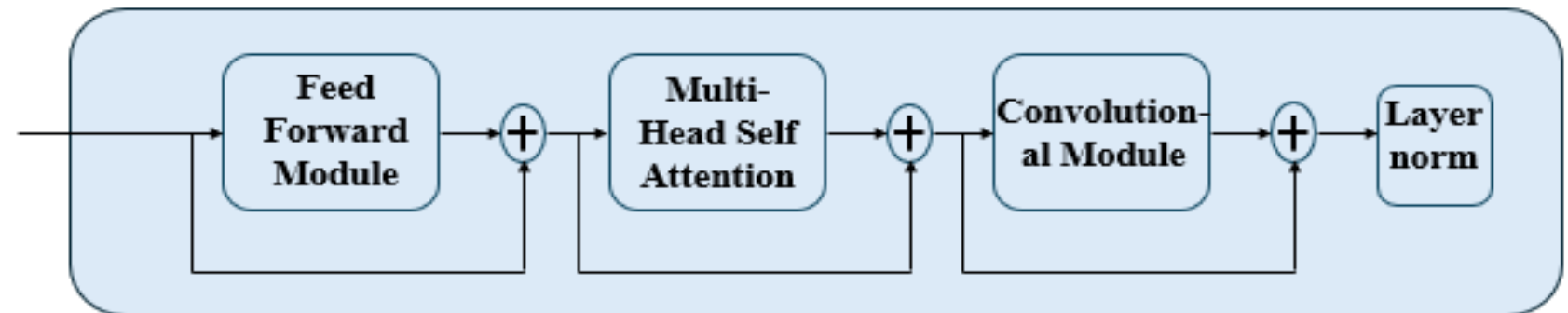
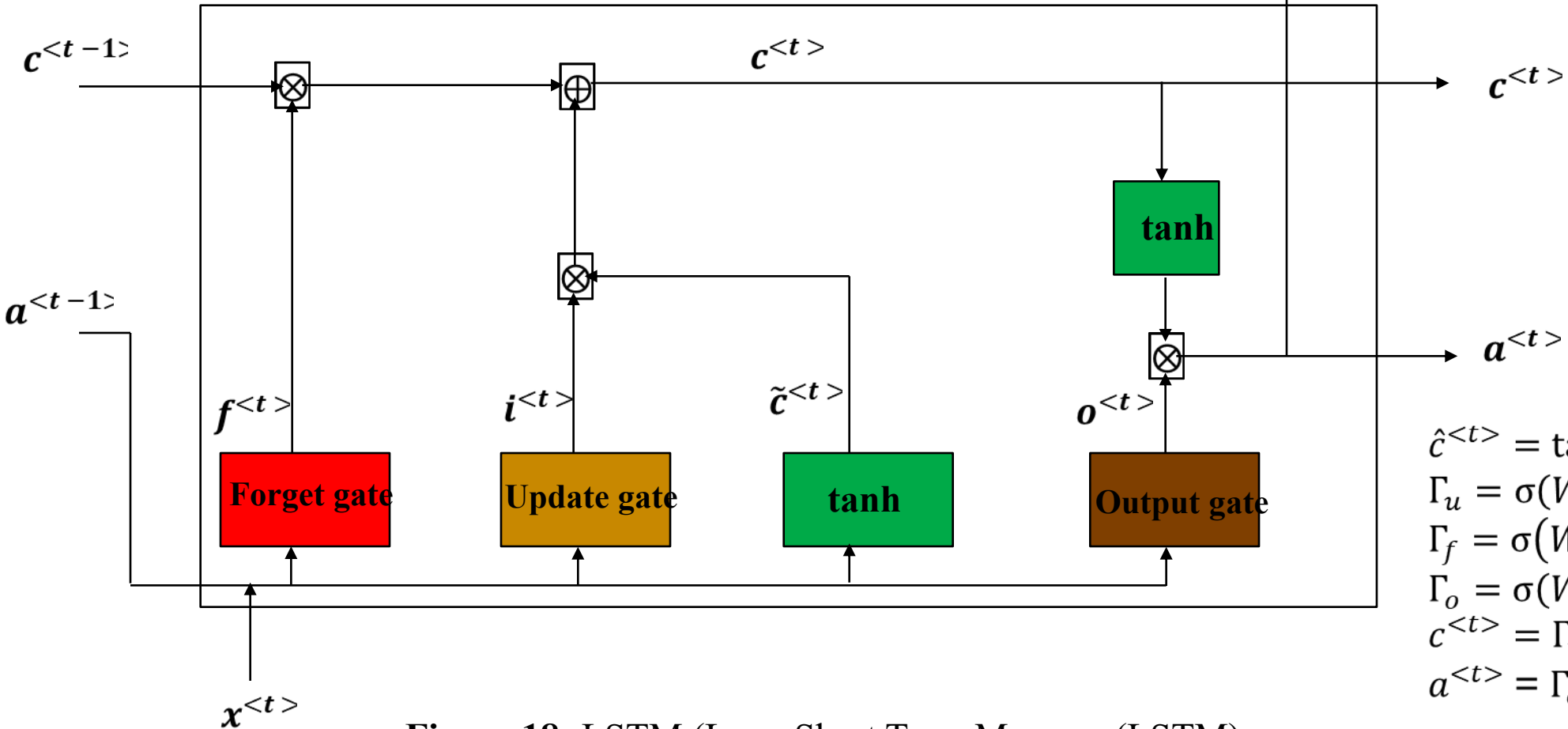


Figure 17: Conformer Block with Altered Feed Forward Module

J. Decoder



$$\begin{aligned}\hat{c}^{<t>} &= \tanh(W_c[a^{<t-1>}, x^{<t>}] + b_c) \\ \Gamma_u &= \sigma(W_u[a^{<t-1>}, x^{<t>}] + b_u) \\ \Gamma_f &= \sigma(W_f[a^{<t-1>}, x^{<t>}] + b_f) \\ \Gamma_o &= \sigma(W_o[a^{<t-1>}, x^{<t>}] + b_o) \\ c^{<t>} &= \Gamma_u * \hat{c}^{<t>} + \Gamma_f * c^{<t-1>} \\ a^{<t>} &= \Gamma_o * \tanh c^{<t>}\end{aligned}$$

Figure 18: LSTM (Long Short Term Memory (LSTM))

J. Decoder

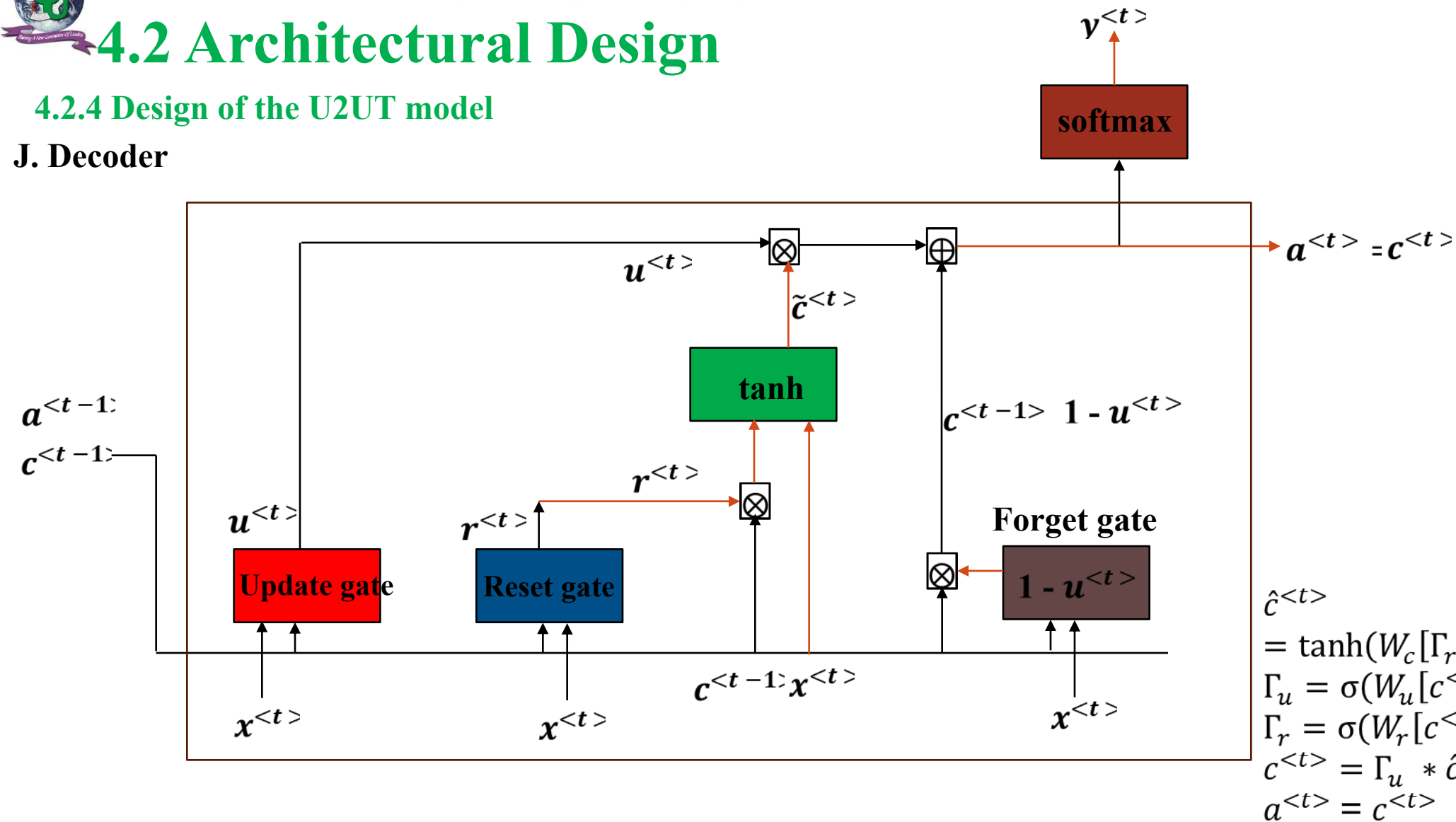


Figure 19: LSTM (Gated Recurrent Unit (GRU))

4.2 Architectural Design

4.2.4 Design of the U2UT model

K. Decoder Block

- ❖ This comprises of groups of LSTM decoder.
- ❖ It was called a Phoneme decoder block to maintain the name in the vanilla Translatotron2.

L. Cross-Attention

- ❖ Multi-Head Cross Attention (MHCA) is meant to find the relationship (mapping) between the source English embedded discretised input and target Yoruba discrete values.

M. Synthesiser

1. Duration Predictor

The Duration Predictor can be represented in equation (1) as:

$$DurationPredictor_{out} = Linearlayer\left(ReLU\left(LinearLayer(x)\right)\right) \quad (1)$$

2. Upsampler

Its function was to upsample or project each hidden state to a factor specified by the Duration Predictor.

N. Inference Decoding

4.2 Architectural Design

4.2.4 Design of the U2UT model

J. Decoder (Transformer Decoder)

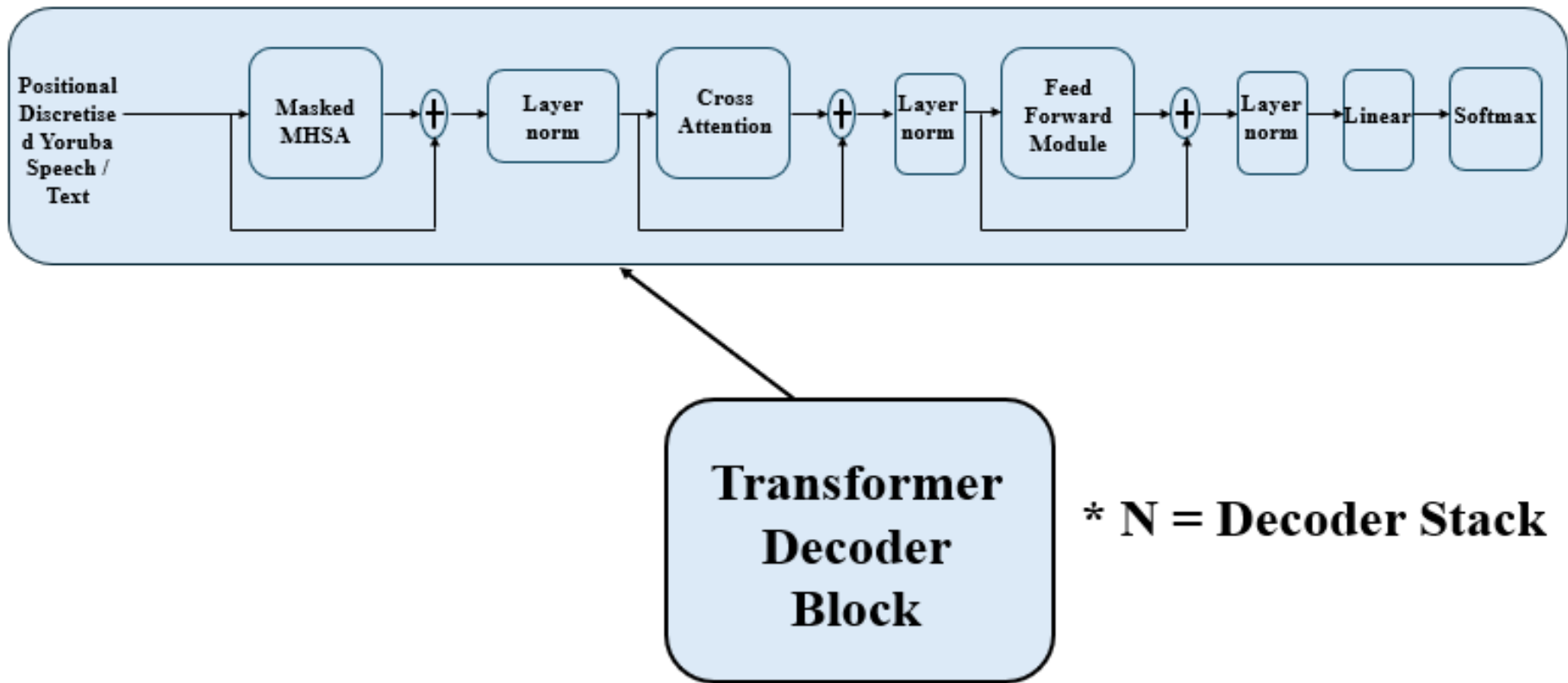


Figure 20: Transformer Decoder

4.2 Architectural Design

4.2.4 Design of the U2UT model

O. U2UT Model

- The U2UT model was developed, which is an integration of all the above sub-modules including the conformer encoder block, with MHSA and positional encoding, phoneme decoder block, cross-attention, and synthesiser with duration predictor and upsampler.
- The U2UT model was then integrated with unit HiFi-GAN vocoder.

4.2.5 U2UT Model Training Experiments

A. U2UT Model Training Experiments with 200 section of General Domain of UI_3

- ❑ Using NVIDIA GeForce RTX 4060 GPU, with Windows 11 operating system, the virtual environment was activated on the Ubuntu with Windows Subsystem for Linux. 200 audio audios from speaker 000 was selected and augmented using to scale the parallel audios to 1400 audios.
- ❑ At start, a dataset sample of size 190 was chosen randomly and partitioned into 158 and 32 parallel audios of English and Yoruba for training and validation respectively.
- ❑ Both checkpoint files and best models were saved.
- ❑ Validation accuracy, as well as training and validation losses were monitored.
- ❑ After over 2 weeks, the dataset was scaled to 336 parallel audios of English and Yoruba with distribution as 266 for training and 70 for validation.
- ❑ This was run for over 2 weeks after which the data was scaled to the full size of 1400 of the augmented audios. The dataset was partitioned into 1120 and 280 for training and validation respectively.
- ❑ Finally, the training commenced with full dataset of 1400 parallel audios and training ran for another 2 weeks

4. METHODOLOGY

4.2 Flow Chart of the Proposed Model

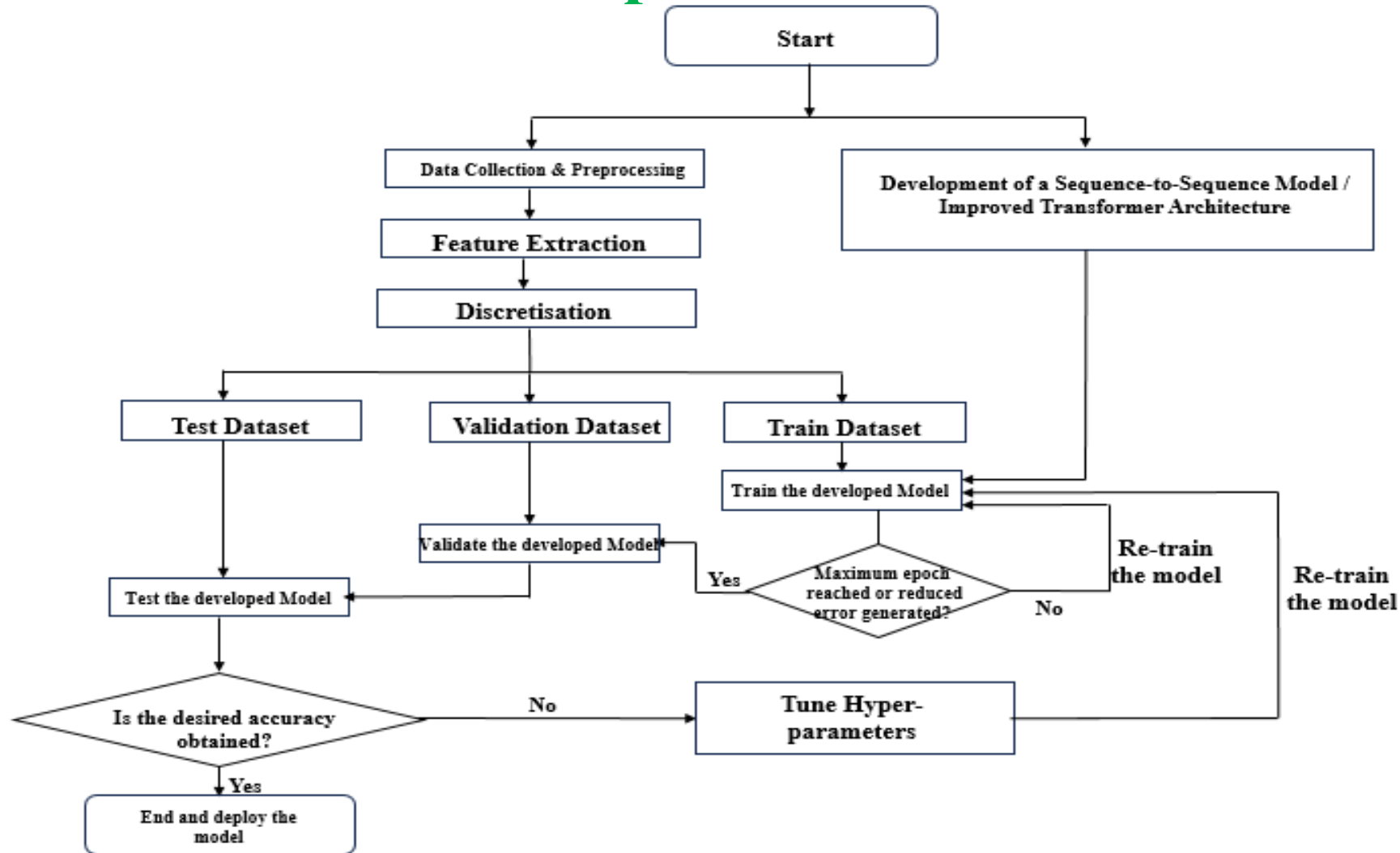


Figure 21: The Flowchart of the proposed Model

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

1. Target Discrete Units Reconstructions (Cont.)

180: Ati niwọn igba ti awọn wọnyi ti o kẹhin ti ọpo gangan nipasẹ ounjẹ, si erogba, nitrogen, ati hydrogen ti ounjẹ naa.

original directly from Hubert for sample 180

[17, 17, 17, 296, 392, 392, 296, 296, 442, 201, 363, 296, 201, 201, 201, 201, 201, 201, 201, 201, 201, 201, 201, 237, 237, 237, 237, 237, 237, 201, 201, 237, 237, 237, 237, 237, 237, 237, 237, 442, 442, 442, 442, 237, 237, 237, 237, 40, 40, 442, 442, 237, 305, 305, 305, 305, 201, 1, 1, 1, 21, 21, 21, 21, 21, 21, 21, 21, 101, 101, 101, 101, 101, 228, 289, 289, 82, 82, 83, 145, 111, 240, 240, 35, 478, 162, 232, 172, 172, 224, 267, 267, 116, 10, 10, 10, 398, 398, 398]

Constructed from evaluation at epoch 434 for sample 180

[17, 17, 17, 296, 392, 392, 296, 296, 442, 201, 363, 296, 201, 201, 201, 201, 201, 201, 201, 201, 201, 201, 201, 237, 237, 237, 237, 237, 237, 201, 201, 237, 237, 237, 237, 237, 237, 237, 237, 442, 442, 442, 442, 237, 237, 237, 237, 40, 40, 442, 442, 237, 305, 305, 305, 305, 201, 201, 1, 1, 21, 21, 21, 21, 21, 21, 21, 21, 101, 101, 101, 101, 101, 228, 289, 289, 82, 82, 83, 145, 111, 240, 240, 35, 478, 162, 232, 172, 172, 224, 267, 267, 116, 10, 10, 10, 398, 398, 398]

Constructed from evaluation at epoch 497 for sample 180

[17, 17, 17, 296, 392, 392, 296, 296, 442, 201, 363, 296, 201, 201, 201, 201, 201, 201, 201, 201, 201, 201, 201, 237, 237, 237, 237, 237, 237, 201, 201, 237, 237, 237, 237, 237, 237, 237, 237, 442, 442, 442, 442, 237, 237, 237, 237, 40, 40, 442, 442, 237, 305, 305, 305, 305, 201, 1, 1, 1, 21, 21, 21, 21, 21, 21, 21, 21, 101, 101, 101, 101, 101, 228, 289, 289, 82, 82, 83, 145, 111, 240, 240, 35, 478, 162, 232, 172, 172, 224, 267, 267, 116, 10, 10, 10, 398, 398, 398]

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

2. Continuous loss in training between epoch 530 to 540 (37,100 to 37800 steps)

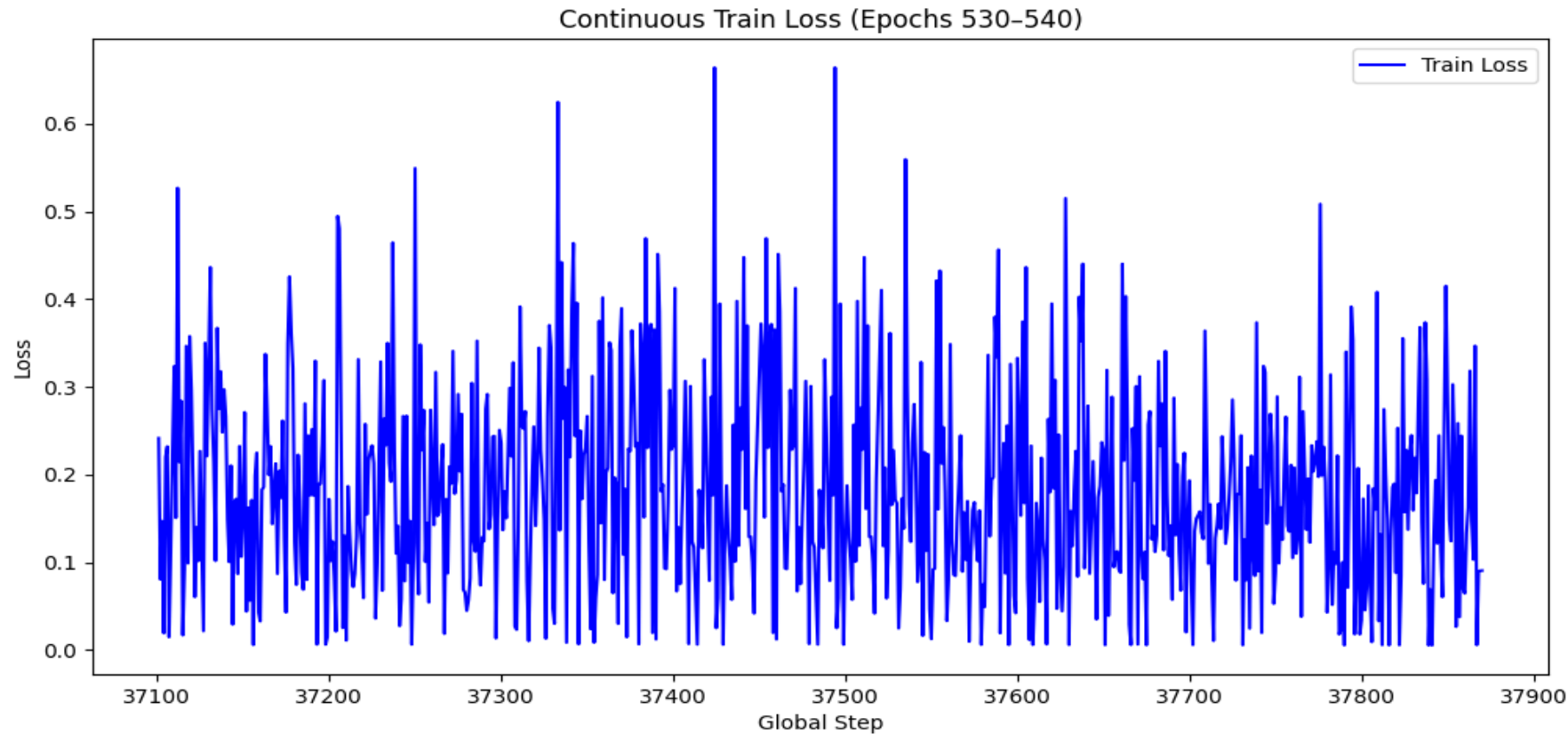


Figure 22: Train Loss at Epoch 530 to 540

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

3. Continuous loss in validation between epochs 530 to 540

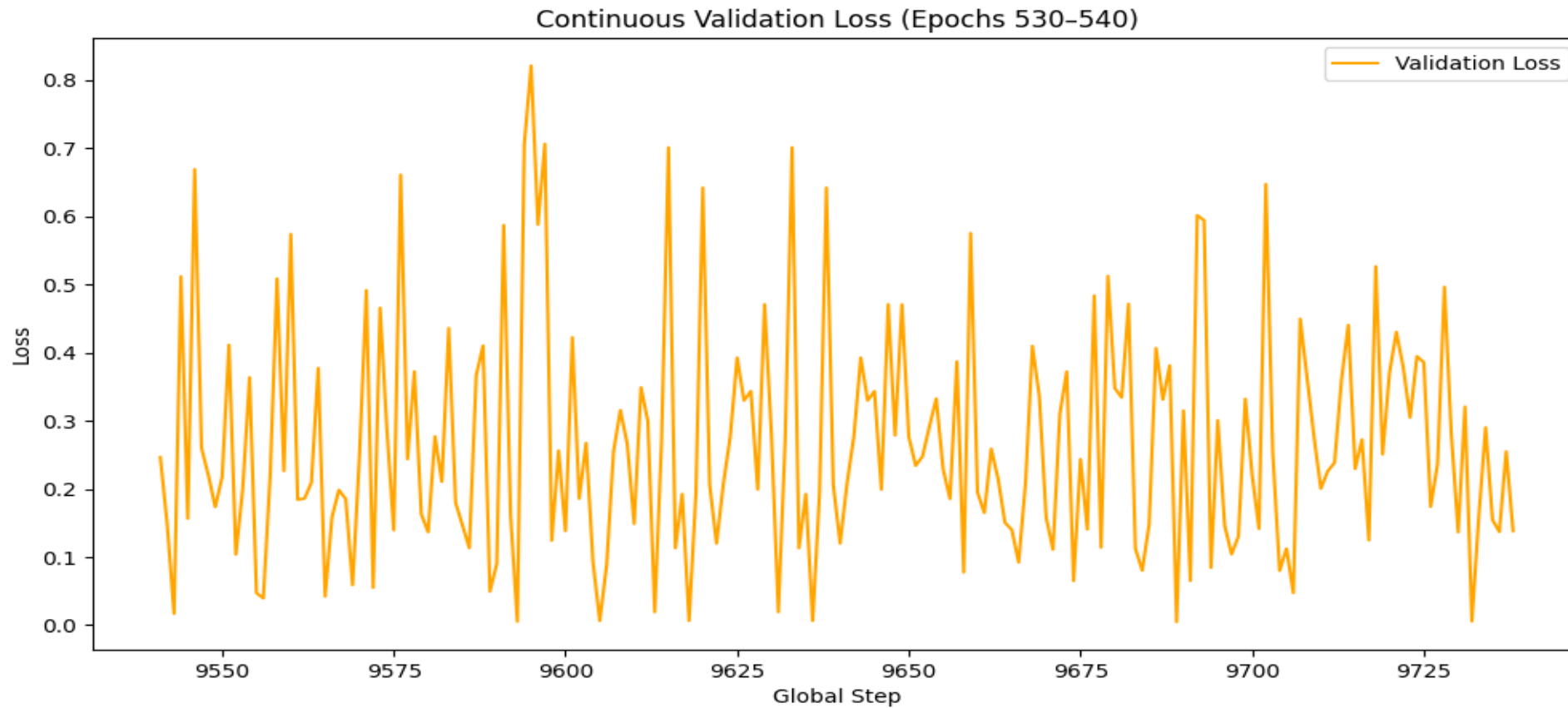


Figure 23: Validation Loss at Epoch 530 to 540

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

4. Metrics parameters at epoch 524

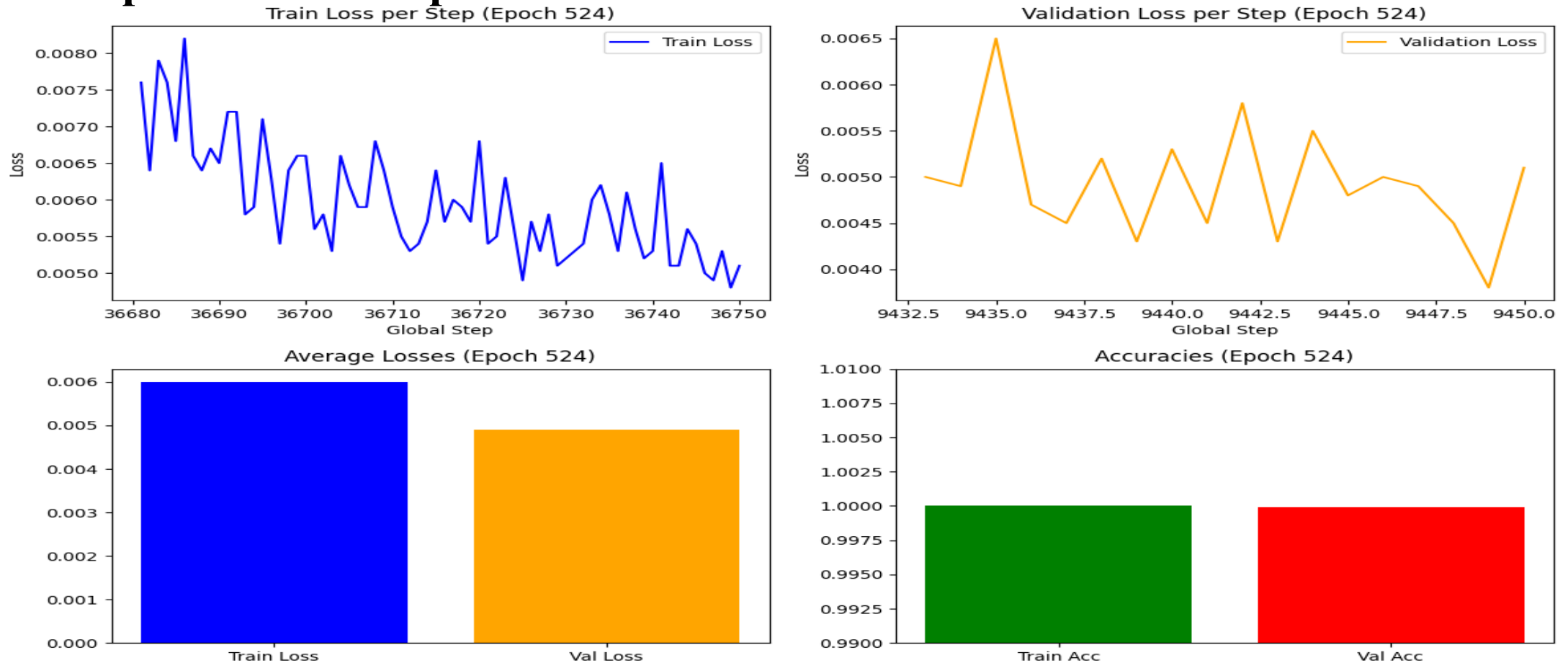


Figure 24: Training Metrics at Epoch 524

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

5. Metrics parameters at epoch 525

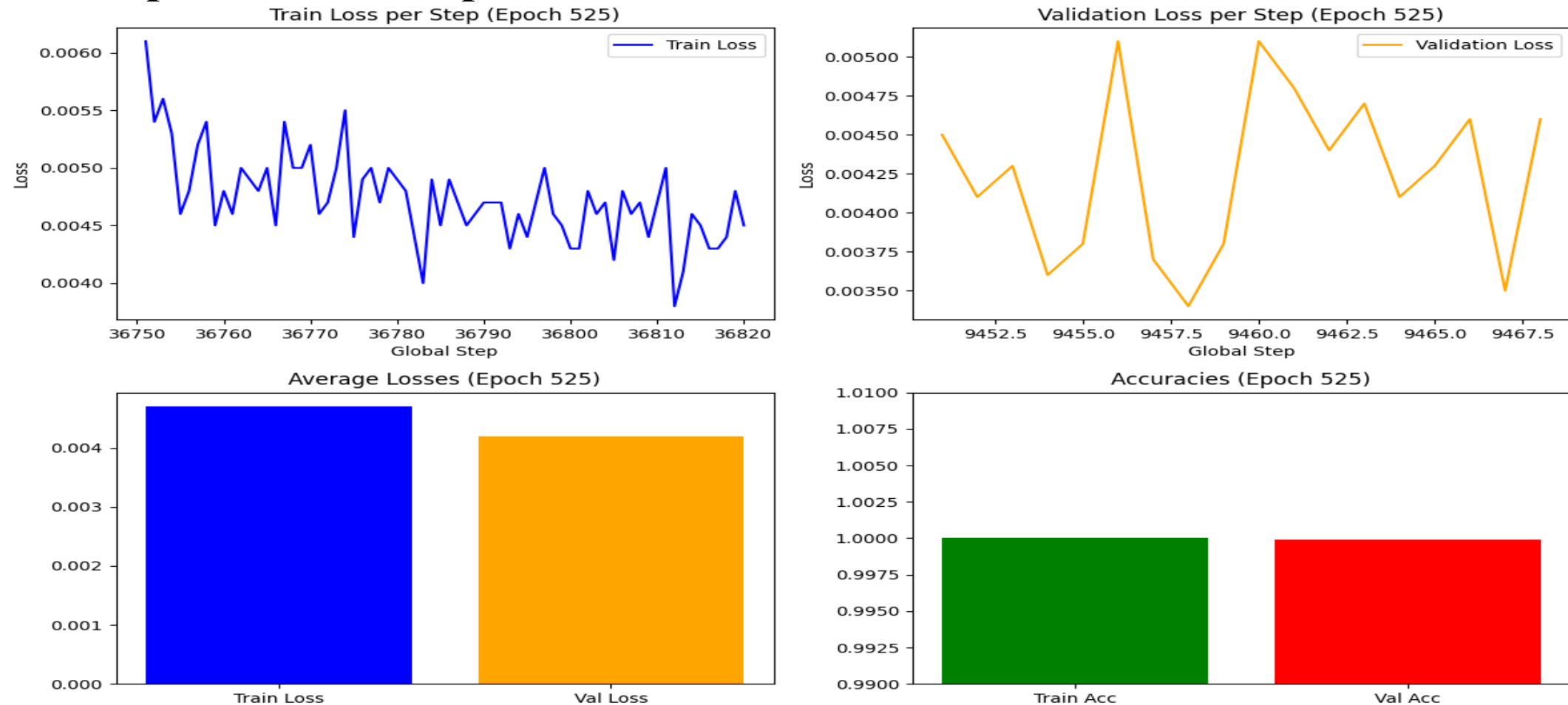


Figure 25: Training Metrics at Epoch 525

5. RESULTS AND DISCUSSION

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

6. NCAIR ASR results of the ground truth, original audios' texts, and selected audios from U2UT model

Table 5: ASR Outputs in Comparison to the Ground Truth Texts of Four Selected Audio Samples

Selected Samples\Features	Ground Truth	Original Audio Fed Directly to ASR	From U2UT model
Sample 184 @ 504 with 1400 audios	Şugbon, niwon bi metamorphosis ti awon eya ti a seto si n tesiwaju die sii laiara, aipe awon nkan yen yoo waye.	òşògbón ni wón bí mēta mō pòòsì ti àwón èyà tí a şèdò sí èfēsìwájú díè sí láì yàrá àì pé àwón òkàn yen yóò wáyé	mi ò mō mi mēpamō fòòsì sí àiyan èyà tí a şètò sí ifēsìwà tí jè sí láìní ara àipè òpò yen yóò wáyé
Sample 186 @ 484 with 1400 audios	Pupò kere si erogba ati hydrogen ni fòmu ti a şe deede fun ilana ategun ju ni ibamu si ategun ti o gba ninu edoforo.	púpò kékeré şe irùgbà àti àjójù ni fòòmù tí a şèlédé fún ilànà àtègùn jù ní ìbámù sí àtègùn tí ó gbà nínú èdòfò	àwón òkèké rè sí irùbà àti agbòşemi ni fòòmù tí a şe dínđín fì maa tēgbón gbògbé ìgbámù sí àtewò tí irùbà ìgbé egbé.
Sample 184 @ 455 with 1400 audios	Şugbon, niwon bi metamorphosis ti awon eya ti a seto si n tesiwaju die sii laiara, aipe awon nkan yen yoo waye.	òşògbón ni wón bí mēta mō pòòsì ti àwón èyà tí a şèdò sí èfēsìwájú díè sí láì yàrá àì pé àwón òkàn yen yóò wáyé	mi ò mō, mi mēpamō fòòsì sí àiyan èyà tí a şètò sí ifēsìwà tí jè sí láìní ara àipè òpò yen yóò wà yéyè
Sample 187 @ 467 with 1400 audios	Ati pe yoo je dandan fun ise ti ategun, kii se aipe erogba ati hydrogen ti a pese lati orisun miiran.	àti pé yóò jè dọdọ fún işè tí a tēgùn kí a kí şe àipè-irùgbà àti àìjírìní tí a pèsè láti orísùmiràn.	ọti pe irúdì ìbòdó fún işè tí a tēgùn kí a şíkí şe àipè irùbà àti àìjíríà tí a pè sọlá tí wón isún miiran.

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

7. BLEU scores of original audios and corresponding selected outputs of U2UT model

Table 6: BLEU Score of Four Selected Audios with Original NCAIR ASR Texts Outputs

Selected Samples	Original Audio Fed Directly to ASR	From U2UT model
Sample 184 @ 504 with 1400 audios	1.90	1.79
Sample 186 @ 484 with 1400 audios	2.05	3.44
Sample 184 @ 455 with 1400 audios	1.90	1.90
Sample 187 @ 467 with 1400 audios	3.92	2.04

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

8. BLEU scores (normalized texts) of original audios and corresponding selected outputs of U2UT model

Table 7: BLEU Score of Five Selected Audios with Normalised NCAIR ASR Texts Outputs

Selected Samples	Original Audio Fed Directly to ASR		From U2UT model	
	BLEU Scores	CER	BLEU Scores	CER
Sample 184 @ 504 with 1400 audios	20.02	0.40	20.83	0.59
Sample 186 @ 484 with 1400 audios	57.05	0.14	16.36	0.59
Sample 184 @ 455 with 1400 audios	20.02	0.40	17.27	0.60
Sample 163 @ 451 with 1400 audios	53.69	0.29	2.35	0.64
Sample 187 @ 467 with 1400 audios	27.74	0.44	9.01	0.37

5. RESULTS AND DISCUSSION

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

9. Normalised NCAIR ASR results

Table 8: ASR Normalised Outputs in Comparison to the Ground Truth Normalised Texts of Four Selected Audio Samples

Selected Samples\Features	Ground Truth	Original Audio Fed Directly to ASR	From U2UT model
Sample 184 @ 504 with 1400 audios	sugbon niwon bi metamorphosis ti awon eya ti a seto si n tesiwaju die sii laiyara aipe awon nkan yen yoo waye.	osogbon ni won bi meta mo poosi ti awon eya ti a sedo si efesiwaju die si lai yara ai pe awon okan yen yoo waye.	mi o mo mi mepamo foosi si aiyan eya ti a seto si ifesiwa ti je si laini ara aipe opo yen yoo waye
Sample 186 @ 484 with 1400 audios	pupo kere si erogba ati hydrogen ni foomu ti a se deede fun ilana ategun ju ni ibamu si ategun ti o gba ninu edoforo	pupo kekere se irugba ati ajoju ni foomu ti a seledede fun ilana ategun ju ni ibamu si ategun ti o gba ninu edofoo	awon okeke re si iruba ati agbosemi ni foomu ti a se dindin fi maa tegbon gbogbe igbamu si atewo ti iruba igbe egbe
Sample 184 @ 455 with 1400 audios	sugbon niwon bi metamorphosis ti awon eya ti a seto si n tesiwaju die sii laiyara aipe awon nkan yen yoo waye.	osogbon ni won bi meta mo poosi ti awon eya ti a sedo si efesiwaju die si lai yara ai pe awon okan yen yoo waye.	mi o mo mi mepamo foosi si aiyan eya ti a seto si ifesiwa ti je si laini ara aipe opo yen yoo wa yeye
Sample 187 @ 467 with 1400 audios	ati pe yoo je dandan fun ise ti ategun kii se aipe erogba ati hydrogen ti a pese lati orisun miiran.	ati pe yoo je dodo fun ise ti a tegun ki a ki se aipeirugba ati aijiriini ti a pese lati orisumiran.	oti pe irudi ibodo fun ise ti a tegun ki a siki se aipe iruba ati aijiria ti a pe sola ti won isun miiran

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

10. PESQ and STOI scores of selected outputs of U2UT model

Table 9: PESQ and STOI of Seven Selected Audio Outputs

Selected Samples	U2UT model	
	PESQ	STOI
Sample 180 @ 478 with 1400 audios	2.58	0.13
Sample 184 @ 565 with 1400 audios	1.79	0.18
Sample 184 @ 451 with 1400 audios	1.64	0.17
Sample 187 @ 467 with 1400 audios	1.67	0.13
Sample 180 @ 434 with 1400 audios	1.09	0.55
Sample 183 @ 451 with 1400 audios	1.17	0.46
Sample 187 @ 467 with 1400 audios	1.11	0.58

5.1 U2UT Model Training Experiments with 200 audios of General Domain of UI_3

11. DNSMOS scores of selected outputs of U2UT model

Table 10: DNSMOS of Randomly Selected Audios and U2UT model outputs

Selected Samples	Original Audio Fed Directly to ASR	From U2UT model
Sample 189 @ 520 with 1400 audios	1.90	3.00
Sample 190 @ 524 with 1400 audios	2.33	3.12
Sample 163 @ 451 with 1400 audios	2.92	3.20
Sample 187 @ 451 with 1400 audios	2.57	2.64
Sample 164 @ 536 with 1400 audios	2.86	2.40
Sample 175 @ 526 with 1400 audios	2.60	1.97
Sample 197 @ 478 with 1400 audios	2.27	2.04
Sample 185 @ 526 with 1400 audios	2.78	2.09

6.1 Overall Statistics (03/03/2025 – 09/07/2026)

Table 11: Overall Statistics [230,684 audios (516.98 hrs.); Speakers (581 + 8); # 6,024,124.9 – 09/07/2026]

S/N	Institutions	No of participants Projected	No of Registered participants	No of participants that completed	Amount Spent
1	UNILAG	100	Whatsapp – 50; Online - 38	22 (PH: 11; GEN: 11) + 5	# 434,124.90
2	UI	100	Whatsapp – 159; Online – 143, New Whatsapp – 194	84 (PH: 50; GEN: 34) + 2	# 860,000.00
3	FUNAAB	100	Whatsapp – (99+40); Online – 59	45 (PH: 20; GEN: 25)	# 450,000.00
4	LASU	100	Whatsapp – 90; Online - 88	5 (PH: 1; GEN: 4)	# 50,000.00
5	OAU	100	Whatsapp – None; Online – None	None	None
6	UI_2 + UI_3	100	Whatsapp – 200+; Online – not yet verified	96 (PH: 57; GEN: 39) 99 (PH: 96; GEN: 3)	# 960,000.00 # 990,000.00
7	UI_BATCH2	100	Whatsapp – 180;	106 (PH: 45; GEN: 61)	# 1,060,000.00
8	UI_BATCH2_2	100	Whatsapp – 180+;	64 (PH: 62; GEN: 02)	# 640,000.00

6.1 Overall Statistics (03/03/2025 – 09/07/2026)

Table 11: Overall Statistics [230,684 audios (516.98 hrs.); Speakers (581 + 8); # 6,024,124.9 – 09/07/2026]

S/N	Institutions	No of participants Projected	No of Registered participants	No of participants that completed	Amount Spent
9	UI_6	100	Whatsapp – 180+;	58 (PH: 29; GEN: 29)	# 580,000.00

6.2 University of Lagos

Table 12: UNILAG Statistics (22 + 5 Speakers)

1	num_audios_src	4253
2	num_audios_tgt	4253
3	total_num_audios (src+tgt)	8506
4	len_audio(time)_src (secs)	28961.746 (8 hrs.)
5	len_audio(time)_tgt (secs)	29151.519 (8 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	58113.26 (16 hrs.)
7	num_words_src	40404
8	num_unique_words_src*	16825
9	num_words_tgt	44247
10	num_unique_words_tgt*	16098
11	num_of_characters_src	193862
12	num_of_characters_tgt	171099
13	num_lines_of_text_src	4253
14	num_lines_of_text_tgt	4253
15	num_lines_of_text_total (src + tgt)	8506

6.3 University of Ibadan, UI

Table 13: UI Statistics (84 + 2 Speakers)

1	num_audios_src	16,545
2	num_audios_tgt	16,545
3	total_num_audios (src+tgt)	33090
4	len_audio(time)_src (secs)	114240.87 (31.73 hrs.)
5	len_audio(time)_tgt (secs)	127335.19 (35.37 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	241576.08 (67.10 hrs.)
7	num_words_src	164230
8	num_unique_words_src*	64501
9	num_words_tgt	178084
10	num_unique_words_tgt*	64869
11	num_of_characters_src	781462
12	num_of_characters_tgt	686416
13	num_lines_of_text_src	16,545
14	num_lines_of_text_tgt	16,545
15	num_lines_of_text_total (src + tgt)	33090

6.3 University of Ibadan, UI (Cont.)

Table 14: UI_2 Statistics (96 Speakers)

1	num_audios_src	19,089
2	num_audios_tgt	19,089
3	total_num_audios (src+tgt)	38,178
4	len_audio(time)_src (secs)	112,449.6225 (31.24 hrs.)
5	len_audio(time)_tgt (secs)	129,585.15547 (36.00 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	242,034.77797 (67.24 hrs.)
7	num_words_src	160,992
8	num_unique_words_src*	54,432
9	num_words_tgt	178,327
10	num_unique_words_tgt*	47,971
11	num_of_characters_src	899,072
12	num_of_characters_tgt	785,255
13	num_lines_of_text_src	19,089
14	num_lines_of_text_tgt	19,089
15	num_lines_of_text_total (src + tgt)	38,178

6.3 University of Ibadan, UI (Cont.)

Table 15: UI_3 Statistics (99 Speakers)

1	num_audios_src	19,670
2	num_audios_tgt	19,670
3	total_num_audios (src+tgt)	39,340
4	len_audio(time)_src (secs)	143,350.773567 secs (39.82 hrs.)
5	len_audio(time)_tgt (secs)	171,491.200867 secs (47.64 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	314,841.958434 secs (87.46 hrs.)
7	num_words_src	223,063
8	num_unique_words_src*	70,973
9	num_words_tgt	2261,647
10	num_unique_words_tgt*	63,274
11	num_of_characters_src	1,293,412
12	num_of_characters_tgt	1,182,595
13	num_lines_of_text_src	19,670
14	num_lines_of_text_tgt	19,670
15	num_lines_of_text_total (src + tgt)	39,340

6.3 University of Ibadan, UI (Cont.)

Table 16: UI_BATCH2 (UI_4) Statistics (108 Speakers)

1	num_audios_src	21,525
2	num_audios_tgt	21,525
3	total_num_audios (src+tgt)	43,050
4	len_audio(time)_src (secs)	175,867.93564 secs (48.85 hrs.)
5	len_audio(time)_tgt (secs)	215,312.30856 secs (59.81 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	391,180.2442 secs (108.66 hrs.)
7	num_words_src	311,376
8	num_unique_words_src*	116,233
9	num_words_tgt	362,813
10	num_unique_words_tgt*	120,394
11	num_of_characters_src	1,563,223
12	num_of_characters_tgt	1,502,789
13	num_lines_of_text_src	21,525
14	num_lines_of_text_tgt	21,525
15	num_lines_of_text_total (src + tgt)	43,050

6.3 University of Ibadan, UI (Cont.)

Table 17: UI_BATCH2_2 (UI_5) Statistics (64 Speakers)

1	num_audios_src	12,751
2	num_audios_tgt	12,751
3	total_num_audios (src+tgt)	25,502
4	len_audio(time)_src (secs)	105,419.5101 secs (29.28 hrs.)
5	len_audio(time)_tgt (secs)	127,343.225933 secs (35.37 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	232,762.736033 secs (64.65 hrs.)
7	num_words_src	149,088
8	num_unique_words_src*	56,638
9	num_words_tgt	182,877
10	num_unique_words_tgt*	55,459
11	num_of_characters_src	865,051
12	num_of_characters_tgt	845,926
13	num_lines_of_text_src	12,751
14	num_lines_of_text_tgt	12,751
15	num_lines_of_text_total (src + tgt)	25,502

6.3 University of Ibadan, UI (Cont.)

Table 18: UI_BATCH2_3 (UI_6) Statistics (58 Speakers)

1	num_audios_src	11,575
2	num_audios_tgt	11,575
3	total_num_audios (src+tgt)	23,150
4	len_audio(time)_src (secs)	86,637.22899 secs (24.07 hrs.)
5	len_audio(time)_tgt (secs)	107,268.75893 secs (29.80 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	193,905.99239 secs (53.87 hrs.)
7	num_words_src	143,390
8	num_unique_words_src*	55,554
9	num_words_tgt	162,547
10	num_unique_words_tgt*	51,407
11	num_of_characters_src	719,141
12	num_of_characters_tgt	669,549
13	num_lines_of_text_src	11,575
14	num_lines_of_text_tgt	11,575
15	num_lines_of_text_total (src + tgt)	23,150

6.4 Federal University of Agriculture Abeokuta, FUNAAB

Table 19: FUNAAB Statistics (45 Speakers)

1	num_audios_src	8,953
2	num_audios_tgt	8,953
3	total_num_audios (src+tgt)	17,906
4	len_audio(time)_src (secs)	78,831.28686 (21.90 hrs)
5	len_audio(time)_tgt (secs)	89,138.5185 (24.76 hrs)
6	len_audio(time)_total (src + tgt) (secs)	167,969.80536 (46.66 hrs)
7	num_words_src	102,197
8	num_unique_words_src*	36,157
9	num_words_tgt	111,779
10	num_unique_words_tgt*	37,566
11	num_of_characters_src	500,223
12	num_of_characters_tgt	445,647
13	num_lines_of_text_src	8,953
14	num_lines_of_text_tgt	8,953
15	num_lines_of_text_total (src + tgt)	17,906

6 OUTCOMES OF METHOD 1: DATA CURATION FEEDBACK

6.5 Lagos State University, LASU

Table 20: LASU Statistics (5 Speakers)

1	num_audios_src	981
2	num_audios_tgt	981
3	total_num_audios (src+tgt)	1,962
4	len_audio(time)_src (secs)	8,633.685334 (2.39 hrs.)
5	len_audio(time)_tgt (secs)	10,137.173334 (2.81 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	18,770.858664 (5.21 hrs.)
7	num_words_src	12,373
8	num_unique_words_src*	4,691
9	num_words_tgt	13,016
10	num_unique_words_tgt*	5,012
11	num_of_characters_src	56,754
12	num_of_characters_tgt	50,552
13	num_lines_of_text_src	981
14	num_lines_of_text_tgt	981
15	num_lines_of_text_total (src + tgt)	1,962

6.6 Percentage Collected

Table 21: PERCENTAGE OF NUMBER OF AUDIOS COLLECTED

NUMBER OF AUDIOS PROJECTED = 200,000; OBTAINED PER LANGUAGE: 115,342 AUDIOS

	UNILAG	UI	FUNAAB	LASU	UI_2	UI_3	UI_Batch2 (UI_4)
Total number of EN	4,253 /18,181.81 (23.39%)	16,545 /18,181.81 (91.00%)	8,953/18,181.81 (49.24%)	981/18,181.81 (5.40%)	19,089/18,181.81 (104.99%)	19,670/18,181.81 (108.19%)	21,525/18,181.81 (118.39%)
Total number of YOR	4,253 /18,181.81 (23.39%)	16,545 /18,181.81 (91.00%)	8,953/18,181.81 (49.24%)	981/18,181.81 (5.40%)	19,089/18,181.81 (104.99%)	19,670/18,181.81 (108.19%)	21,525/18,181.81 (118.39%)
Overall %	$4253 + 16545 + 8,953 + 981 + 19,089 + 19,670 + 21,525 + 12,751 + 11,575 = 115,342 / 200,000 = 57.67\%$						

6 OUTCOMES OF METHOD 1: DATA CURATION FEEDBACK

6.6 Percentage Collected (Cont.)

Table 21: PERCENTAGE OF NUMBER OF AUDIOS COLLECTED (CONT.)

NUMBER OF AUDIOS PROJECTED = 200,000; OBTAINED PER LANGUAGE: 115,342 AUDIOS

	UI_Batch2_2 (UI_5)	UI_6					
Total number of EN	12,751/18,181.81 (70.13%)	11,575/18,181.81 (63.66%)					
Total number of YOR	12,751/18,181.81 (70.13%)	11,575/18,181.81 (63.66%)					
Overall %	$4253 + 16545 + 8,953 + 981 + 19,089 + 19,670 + 21,525 + 12,751 + 11,575 = 115,342 / 200,000 = 57.67\%$						

6 OUTCOMES OF METHOD 1: DATA CURATION FEEDBACK

6.6 Percentage Collected (Cont.)

Table 22: PERCENTAGE OF HOURS OF AUDIOS COLLECTED

HOURS OF AUDIOS PROJECTED = 1,000 hrs.;

OBTAINED: 237.32 hrs. (EN), 279.66 hrs. (YOR), TOTAL = 516.98 hrs.

	UNILAG	UI	FUNAAB	LASU	UI_2	UI_3	UI_Batch 2 (UI_4)
Total hours of EN	8.04	31.73	21.90	2.39	31.24	39.82	48.85
Total hours of YOR	8.10	35.37	24.76	2.81	36.00	47.64	59.81
Total hours (Expected = 90.90 hrs. (9.09%))	16.14 (1.61%)	67.10 (6.71%)	46.66 (4.67%)	5.21 (0.52%)	67.24 (6.72%)	87.46 (8.75%)	108.66 (10.86%)
Overall %	$16.14 + 67.10 + 46.66 + 5.21 + 67.24 + 87.46 + 108.66 + 64.65 + 53.87 = 516.98 / 1,000$ = 51.70 %						

6 OUTCOMES OF METHOD 1: DATA CURATION FEEDBACK

6.6 Percentage Collected (Cont.)

Table 22: PERCENTAGE OF HOURS OF AUDIOS COLLECTED (CONT.)

HOURS OF AUDIOS PROJECTED = 1,000 hrs.;

OBTAINED: 237.32 hrs. (EN), 279.66 hrs. (YOR), TOTAL = 516.98 hrs.

	UI_Batch2_2 (UI_5)	UI_6					
Total hours of EN	29.28	24.07					
Total hours of YOR	35.37	29.80					
Total hours (Expected = 90.90 hrs. (9.09%))	64.65 (6.46%)	53.87 (5.39%)					
Overall %	$16.14 + 67.10 + 46.66 + 5.21 + 67.24 + 87.46 + 108.66 + 64.65 + 53.87 = 516.98 / 1,000$ = 51.70 %						

**FROM 22 + 84 + 45 + 5 + 96 + 99 + 108 + 64 + 58 = 581 DOCUMENTED
SPEAKERS / 1100 OBTAINABLE; % OF SPEAKERS = 52.82 %**

Table 23: Research Budget

S/n	ITEM	COST (\$)	QUANTITY	TOTAL (\$)
1	Speech Data Curation (Google PhD)		10 weeks	10,064.14
3	Conferences (CApIC ACE)	750	2	1,500
4	Publications (CApIC ACE)	2000	2	4,000
	Total			15,564.14

Table 24: Research Timeline Detailing Semester Breakdown of Activities

TASK NO	TASK NAME	START DATE	END DATE	OMEGA SEMESTER , 2023/2024	ALPHA SEMESTE R 2024/2025	OMEGA SEMESTE R 2024/2025	ALPHA SEMESTE R 2025/2026	OMEGA SEMESTE R 2025/2026
1	Development of Proposal	04/01/2024	01/06/2024	Yes				
2	Field Work	09/04/2025	30/03/2026		Yes	Yes	Yes	
3	Methodology	09/04/2025	30/03/2026			Yes	Yes	
4	Results and Discussion	01/01/2026	30/03/2026				Yes	
5	Completion of Report	01/04/2026	27/08/2026					Yes

Table 25: Field Work Breakdown

APRIL-JUNE, 2025	JULY-SEPTEMBER, 2025	OCTOBER, 2025	NOVEMBER-DECEMBER, 2025	JANUARY, 2026	FEBRUARY, 2026	MARCH, 2026
Generation / Curation of dataset Analysis of dataset	Generation / Curation of dataset Understanding of datasets / installation of software packages, modules Analysis of dataset	Generation / Curation of dataset Analysis of dataset	Generation / Curation of dataset Analysis of dataset	Analysis of dataset	Analysis of dataset	Performance analysis of the developed models: Base and Transfer Learning models
Sending out of notification message to intended individual for the task	Familiarisation with the installed software, and collected parallel audio files, as well as dataset available Knowledge of Python programming for deep learning Creation of manifest files	Familiarisation with python modules for the aforementioned models Knowledge of Python programming for deep learning Creation of manifest files	Preparation of dataset for public access Statistical analysis Creation of manifest files	Preparation of dataset for public access Statistical analysis Creation of manifest files	Compare the transfer learning results obtained with that of the BASE MODEL developed. Preparation of dataset for public access Statistical analysis Creation of manifest files	Editing and proof reading of the whole of chapter 3 for corrections, and update Preparation of dataset for public access Statistical analysis Creation of manifest files
Development of web site for collection of data through crowdsourcing	Crowdsourcing and recording in UNILAG, LASU, FUNAAB, and UI	Crowdsourcing and recording in OAU, FUTA, University of Ado-Ekiti, and UNILORIN	Crowdsourcing and recording in University of Lome, Togo, and Univ. of Abomey Calavi, Benin Republic	Crowdsourcing and recording in CU		
Sending out of links for the collection of audio data to intended individual.	Thorough analysis of the dataset for deep understanding, and interpretation using matlab, spss, possibly R	Knowledge of discritisation like the HuBERT, Wav2Vec models and how they are implemented				.

Table 26: Methodology Breakdown

APRIL-JUNE, 2025	JULY-SEPTEMBER, 2025	OCTOBER, 2025	NOVEMBER-DECEMBER, 2025	JANUARY, 2026	FEBRUARY, 2026	MARCH, 2026
Generation / Curation of dataset Development of website Scraping online websites for texts transcripts Use of text-to-text translation Crowd-sourcing individuals for recording	Generation / Curation of dataset Development of models for data preprocessing Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder	Generation / Curation of dataset Development of models for data preprocessing Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder	Generation / Curation of dataset Development of models for data preprocessing Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder	Integration of the developed models into a single base model Train, Validate, Test, and carryout performance evaluation Develop Transfer Learning model Train, Validate, Test, and carryout performance evaluation	Train, Validate, Test, and carryout performance evaluation Develop a prototype Softbot	Integration of developed models to the prototype softbot
Use of Python Flask with HTML, CSS and Vercel server for recording webpage. Thorough search for online media database, newspapers, etc. Use of Google translate Crowdsource individual from 11 institutions to participate in recordings	Same as previous Analysis of data Familiarizations with Python modules for the packages to implement the models. Development of the models mostly taking into consideration the discretization with HuBERT, transformer models.	Same as January Knowledge of the choice of the hyperparameters, for various models of the expected base model. Gain further insight from literatures	Same as February Test each of the developed models. Make a choice of the language translation pretrained model to be utilised for transfer learning	Split the preprocessed discretised dataset into train, validate and test, train the training data using the single developed base, and transfer learning models evaluate using metrics like the MOS, BLEU, scores, precision, recall, F1-score, and ROUGE. [using the FEDGEN platform to run the code]	Train the training data using the single developed base and transfer learning model, evaluate using metrics like the MOS, BLEU, scores, precision, recall, F1-score, and ROUGE. [using the FEDGEN platform to run the code] – For both medical and non medical corpus Ablation studies	Development of another website (Softbot) for interfacing with users. Use of APIs to integrate the model output to the developed web site interface. Sourcing of raters to access the performance of the Softbot

Table 27: Results and Discussion Breakdown

JANUARY, 2026	JANUARY, 2026	FEBRUARY, 2026	FEBRUARY, 2026	MARCH, 2026	MARCH, 2026
Analysis of the results of the various experimental setup for transfer learning and base model for medical and non medical speech corpus	Analysis of the results of the various experimental setup for transfer learning and base model for medical and non medical speech corpus	Updating of the experimental parameters and extraction of new results	Development of write up for the discussion	Development of write up for the discussion	Thorough look through the developed chapter contents for experimental errors and possible corrections.
I shall do both analytical illustration of the results. And graphical illustration if needed.	I shall do both analytical illustration of the results. And graphical illustration if needed.	Based on the already obtained results, there might need to update or enhance the performance of the models via change of hyperparameters of the model.	A coherent discussion of the obtained results will be carried out without bias.	A coherent discussion of the obtained results will be carried out without bias.	Editing and proof reading Checking for experimental hyperparameters erroneously utilised for each of the experiments.
Use of MATLAB, Python for analysis, and interpretation of results	Use of MATLAB, Python for analysis, and interpretation of results	I shall adjust these parameters and report possible improvement if any or as a limitation of the study when certain parameters are adjusted.	Discussion of the findings (analysis, tables, possible chat) will be reported and the whole document of chapter 4 will be integrated as a single file.	Discussion of the findings (analysis, tables, possible chat) will be reported and the whole document of chapter 4 will be integrated as a single file.	Drawing the summary of the chapter.

Table 28: Completion of Report

APRIL - JUNE, 2026	JULY-AUGUST, 2026
Merging of all chapters into a single document	Submission of report
Merge the proof read chapter 1, chapter 2, chapter 3, chapter 4, chapter 5, and appendix into a single document	Submission of the completed report for Viva
	Corrections of errors pointed out.

8.5 Summary of Timeline

Table 29: Summary of Research Timeline

S/N	ACTIVITY	YEAR ONE										YEAR TWO										YEAR THREE															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
1	Defining Research Focus																																				
2	Literature Review																																				
3	Developing Methodology																																				
4	Ethical Application																																				
5	Data Collection																																				
6	Data Analysis																																				
7	Proposal																																				
8	Thesis Writing																																				
9	Publication																																				
10	M I L E S T O N E																																				



9. CONCLUSION

- **The development of speech-to-speech modelling has seen revolution and improvement for high-resourced languages like English, Spanish, French, and German.**
- **However, the African indigenous languages, including the Yoruba language, are still underdeveloped in language translation from another language to Yoruba and vice versa.**
- **This work has curated over 115,000 parallel audios of English and Yoruba (GC-ENYOR-2026 Dataset) and such is to be made available for research in speech processing applications.**
- **U2UT model will be developed for general, public health, and hybrid domain based data, and experiments and ablation studies will be carried out.**
- **Some of the present results are different discrete units reconstructions of the same audios, generations of target discrete units during evaluation.**
- **This work addresses the United Nation Sustainable Development Goals (SDGs) 3-Healthy life and well-being and 10-reducing inequality and the IR4.**

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